Supplementary Material

Size-Agnostic Safe Reinforcement Learning for V2G Charging with a Feasibility-Preserving Safety Layer

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This document supplements the main text. Section S1 proves the formal results, Section S2 documents the modelling in EV2Gym, and Section S3 gives the request pipeline and the perfect-information optima. The compared methods, the training, the tuning and the statistics follow in Section S4, and Section S5 collects further results. Equations, lemmas and figures of the main text are cited with their main-text numbers, and items of this document carry the prefix S. Abbreviations follow the main text on electric vehicle (EV) charging with vehicle-to-grid (V2G) services and photovoltaic (PV) generation at a charge point operator (CPO) station. The main method is the safe attention-based controller (SA-Attn), trained with proximal policy optimization (PPO) and compared with a Deep Sets policy, a multilayer perceptron (MLP), rule-based methods and economic model predictive control (eMPC). Further abbreviations are the linear program (LP), the Karush–Kuhn–Tucker (KKT) conditions, least-laxity-first (LLF) scheduling and conformal prediction (CP). Table S1 lists the main symbols.

S1 Proofs

This section proves the formal results of the main text and states the auxiliary lemmas that the main text cites. All statements hold for every EV separately, and the index i is dropped where no confusion arises. Inequalities are exact in real arithmetic, and the implementation checks them with one tolerance of 10^{-9} in kW or kWh that the unused part of the margin ε_q absorbs.

S1.1 Size independence of the attention policy

Lemma S1 (Equivariance and size independence)**.** *Let ϖ permute the charger tokens and their masks while the global token stays fixed.*

(i) The means μ_θ and standard deviations σ_θ are permutation-equivariant, and the value V_ω is permutation-invariant.

(ii) Outputs of unmasked tokens and the value do not depend on the features or the number of masked tokens.

(iii) The number of parameters does not depend on N.

Proof. Embeddings, layer normalisation, feed-forward blocks and the actor head act row-wise, so they commute with the permutation. For one attention head, the key mask depends only on the key index, so permuting tokens and mask maps the mask matrix M to $PMP^\top$ for the permutation matrix P that fixes the global token. Row-wise softmax satisfies $\mathrm{softmax}(PXP^\top) = P\,\mathrm{softmax}(X)P^\top$, so a masked attention head maps permuted inputs to permuted outputs, and composition gives (i) for the actor. A masked key receives zero weight in every softmax, so its value never enters the output of another token, and together with the row-wise structure this property gives (ii). Attention pooling sums over keys and is therefore invariant, and every weight matrix has a shape fixed by $(F_c, F_g, d_m, n_H, n_L)$ alone, so (iii) holds. The safety layer is permutation-equivariant as well, because its problem is invariant under joint permutation of p, x, l and u and its minimiser is unique. □

Table S1: Main symbols. Units are given in brackets; quantities of EV i carry the index of its charger.

Symbol	Meaning	Symbol	Meaning
N, $\mathcal{N}(t)$, $n(t)$	chargers, occupied chargers, connected EVs	Δt, T	step length [h], horizon [steps]
t_i^{arr}, d_i	arrival step, last controlled step	$\tau_i(t)$	remaining actions, $d_i - t + 1$
$E_i(t)$, E_i^{cap}, $E_i^{\min}$	battery energy, capacity, minimum [kWh]	E_i^{des}, E_i^{req}	desired and admitted energy [kWh]
p_i, $\tilde{p}_i$	commanded and reported power [kW]	g_i, η_i^{c}, η_i^{d}	energy map and efficiencies
$\bar{P}_i(E)$, $\bar{P}_i^{\mathrm{nom}}$	maximum charging power, its nominal value [kW]	$p_i^{\min}(E)$, P_i^{dis}	physical lower limit, discharge power [kW]
P_{tr}, ρ_{tr}	transformer rating, rating per charger [kW]	P^{inf}, P^{pv}	inflexible load, PV generation [kW]
C, C_{rev}, $\tilde{C}$	forward, reverse and tightened forward limit [kW]	ε_q	actuation margin, 0.008 kW
c^{ch}, c^{dis}, c^{thr}	charging, discharge and throughput price [€/kWh]	$\varepsilon_{\mathrm{sat}}$	saturation threshold, 0.001 kW
φ_i, $\Psi_{i,k}$, R_i	one-step full-power map, k-fold map, reachable energy	$B_i(k)$, $\hat{B}_i(k)$	exact and certified thresholds [kWh]
l_i, u_i	lower and upper bound of the safety layer [kW]	χ_i	laxity [steps]
x_i, a_i, P^{ref}	raw action [kW], raw sample, scale 11 kW	$\mathcal{K}(t)$, Π_t	feasible set, projection
λ^*, $\mathcal{F}$	common shift [kW], free set	b	fallback budget [kW]
ς, ς^{tr}, m, κ	state of charge, taper start, multiplier, exponent	$z_0(m)$, $\kappa^*(m)$	exact-delivery constants
ϱ_i^{in}, ϱ_i^{out}	inner and outer taper slope [1/h]	$G_i^{\max}$	rounding allowance [kWh]
J^π, J^{pen}	profit, penalised profit [€]	$J^{\mathrm{pen,LP}}$, $J^{\mathrm{LP,ceil}}$	soft and hard optimum [€]
w_E, w_P, w	evaluation and training penalty weights [€/kWh]	ΔE_i^{sh}	departure shortfall [kWh]
ξ_i, ξ_0	charger and global token	d_m, n_L, n_H	width, layers, heads
θ, ω	actor and critic parameters	β	regulariser weight
γ, λ_{GAE}, ϵ_{clip}	discount, GAE parameter, clip range	n_{mb}, c_{vf}, c_{ent}	minibatch size, loss coefficients
Λ_E, Λ_P, η_Λ	Lagrange multipliers, dual step size	α, n_{cal}	miscoverage level, calibration size
$\hat{q}$, e_j, $\hat{Q}_{1-\alpha}$	quantile predictor, score, conformal quantile	$\hat{d}_i^{\mathrm{lo}}$, τ_i^{lo}	departure bound, remaining steps

The parameter counts are 610,370 for the actor and 676,481 for the critic, 1,286,851 in total for every N. Deep Sets applies a per-token network to each charger token together with the masked mean of all token embeddings and the global token, so the same argument covers it. Its width matches the parameter count of SA-Attn within 10%. The actor of the MLP baseline with 20 slots and two hidden layers of 640 units has 614,440 parameters, a number that grows linearly with the number of slots.

S1.2 Reachable energy and thresholds

Lemma S2 (Reachable energy)**.** *Under A1, Ψ_k is continuous and non-decreasing for every $k \geq 0$. Any admissible power sequence from $E(t)$ gives $E(t+k) \leq \Psi_k(E(t))$, with equality for full-power charging. An EV is individually feasible at step t, meaning that some admissible control meets Equation 4, if and only if $E(t) \geq B(\tau(t))$.*

Proof. A composition of continuous non-decreasing maps is continuous and non-decreasing, so $\Psi_k = \varphi^k$ inherits both properties from A1. For the reach bound we use induction on k, where the case $k = 0$ is trivial. In the induction step, an admissible power $p \leq \bar{P}(E)$ gives $E(t+1) = E + \Delta t\, g(p) \leq E + \Delta t\, g(\bar{P}(E)) = \varphi(E)$ because g is increasing, and the induction hypothesis from $E(t+1)$ with the monotonicity of Ψ_{k-1} gives

$$E(t+k) \leq \Psi_{k-1}\big(E(t+1)\big) \leq \Psi_{k-1}\big(\varphi(E)\big) = \Psi_k(E).$$

Full-power charging gives equality at every step.

For the threshold characterisation we show $\{E \in [E^{\min}, E^{\mathrm{cap}}] : \Psi_k(E) \geq E^{\mathrm{req}}\} = [B(k), E^{\mathrm{cap}}]$ by induction, where the case $k = 0$ is the definition $B(0) = E^{\mathrm{req}}$. The set in Equation 9 is non-empty because $\varphi(E^{\mathrm{cap}}) = E^{\mathrm{cap}} \geq E^{\mathrm{req}} \geq B(k-1)$. In the induction step, $\Psi_k(E) = \Psi_{k-1}(\varphi(E)) \geq E^{\mathrm{req}}$ holds if and only if $\varphi(E) \geq B(k-1)$, and since φ is continuous and non-decreasing, $\{E : \varphi(E) \geq B(k-1)\}$ is a

closed up-set with minimum $B(k)$. By the reach bound, some admissible control meets Equation 4 from $E(t)$ if and only if full-power charging does, that is, if and only if $\Psi_{\tau(t)}(E(t)) \geq E^{\mathrm{req}}$. □

For a certified table, the first two properties of Equation 11 imply $\hat{B} \geq B$ componentwise, because $\varphi(\hat{B}(k)) \geq \hat{B}(k-1) \geq B(k-1)$ gives $\hat{B}(k) \geq B(k)$ by the minimality in Equation 9.

S1.3 Projection

Proof of Proposition 1. (i) If $p \in \mathcal{K}(t)$, then $l \leq p \leq u$ gives $l \leq u$ and $\sum_i l_i \leq \mathbf{1}^\top p \leq \sum_i u_i$, and combined with $-C_{\mathrm{rev}} \leq \mathbf{1}^\top p \leq \tilde{C}$ these inequalities give the four conditions. Conversely, $p(\upsilon) = l + \upsilon(u - l)$ stays in the box for $\upsilon \in [0, 1]$, and $\mathbf{1}^\top p(\upsilon)$ takes every value in $[\sum_i l_i, \sum_i u_i]$, an interval that meets $[-C_{\mathrm{rev}}, \tilde{C}]$ under the four conditions.

(ii) The problem in Equation 12 minimises a strictly convex quadratic over a non-empty polyhedron, so its minimiser exists and is unique, and the KKT conditions are necessary and sufficient. With multipliers $\mu^+, \mu^- \geq 0$ for the aggregate constraints and $\nu^l, \nu^u \geq 0$ for the box, they read

$$p - x + (\mu^+ - \mu^-)\mathbf{1} - \nu^l + \nu^u = 0, \qquad \mu^+(\mathbf{1}^\top p - \tilde{C}) = 0, \qquad \mu^-(\mathbf{1}^\top p + C_{\mathrm{rev}}) = 0,$$
$$\nu_i^l(p_i - l_i) = 0, \qquad \nu_i^u(u_i - p_i) = 0.$$

The aggregate response $\mathcal{S}(\lambda) = \sum_i \mathrm{clip}(x_i - \lambda, l_i, u_i)$ is continuous, non-increasing and piecewise linear, with $\mathcal{S}(\lambda) = \sum_i u_i$ for $\lambda \leq \min_i(x_i - u_i)$ and $\mathcal{S}(\lambda) = \sum_i l_i$ for $\lambda \geq \max_i(x_i - l_i)$. If $\mathcal{S}(0) > \tilde{C} \geq \sum_i l_i$, the intermediate value theorem gives a root of $\mathcal{S}(\lambda) = \tilde{C}$, and every such root is positive; the case $\mathcal{S}(0) < -C_{\mathrm{rev}} \leq \sum_i u_i$ is symmetric with a negative root. Now set $p = \mathrm{clip}(x - \lambda^*, l, u)$, $\mu^+ = \max(\lambda^*, 0)$ and $\mu^- = \max(-\lambda^*, 0)$, and choose the box multipliers coordinate by coordinate. Free coordinates with $l_i < x_i - \lambda^* < u_i$ receive $\nu_i^l = \nu_i^u = 0$, coordinates with $x_i - \lambda^* \leq l_i$ receive $\nu_i^l = l_i - x_i + \lambda^* \geq 0$ and $\nu_i^u = 0$, and the remaining ones receive $\nu_i^u = x_i - \lambda^* - u_i \geq 0$ and $\nu_i^l = 0$. Stationarity then holds coordinate-wise, and complementarity holds because $\lambda^* > 0$ only when $\mathbf{1}^\top p = \tilde{C}$ and $\lambda^* < 0$ only when $\mathbf{1}^\top p = -C_{\mathrm{rev}}$. For $\lambda^* = 0$ the point with $\mathbf{1}^\top p = \mathcal{S}(0)$ is feasible, so in all cases $p = p^*$. If $\lambda_1 < \lambda_2$ are both roots, every coordinate $\mathrm{clip}(x_i - \lambda, l_i, u_i)$ is non-increasing in λ and the sums agree, so all coordinates agree. □

PPO treats the safety layer as part of the environment and never uses its Jacobian. The Jacobian nevertheless describes how the layer passes changes of the raw action to the commanded powers.

Proposition S1 (Jacobian of the projection)**.** *Let $\mathcal{F} = \{i : l_i < x_i - \lambda^* < u_i\}$ be the free set and D the diagonal matrix of its indicators. Wherever $p^* = \Pi_t(x)$ is differentiable, its Jacobian is D if the aggregate constraint is inactive and $D - \mathbf{1}_{\mathcal{F}}\mathbf{1}_{\mathcal{F}}^\top/|\mathcal{F}|$ if it is active with $\mathcal{F} \neq \emptyset$, where $\mathbf{1}_{\mathcal{F}}$ is the indicator vector of $\mathcal{F}$.*

Proof. If the aggregate constraint is inactive, locally $p_i = x_i$ for $i \in \mathcal{F}$ and $p_i \in \{l_i, u_i\}$ is constant otherwise, so the Jacobian is D. With the aggregate constraint active at the target $\vartheta \in \{\tilde{C}, -C_{\mathrm{rev}}\}$, instead, $\sum_{i\in\mathcal{F}}(x_i - \lambda) + \sum_{i\notin\mathcal{F}} p_i = \vartheta$. Hence $\partial\lambda/\partial x_j = \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|$ and $\partial p_i/\partial x_j = \mathbb{1}[i \in \mathcal{F}]\big(\mathbb{1}[i = j] - \mathbb{1}[j \in \mathcal{F}]/|\mathcal{F}|\big)$. If $\mathcal{F} = \emptyset$, the point p is locally constant, and on the null set where p^* is not differentiable, the matrices of the adjacent regions are elements of Clarke's generalised Jacobian. □

The Jacobian is symmetric and idempotent, and as a Euclidean projection onto a convex set, Π_t is non-expansive. On an active row the Jacobian is zero for all clipped coordinates, the zero-gradient effect that motivates treating the layer as part of the environment [1].

S1.4 One-step invariance

The proof sketch of Theorem 1 in the main text and its full proof below both rest on the following lemma.

Lemma S3 (One-step invariance)**.** *Assume A1 and A3, and let $\hat{B}$ satisfy Equation 11. Let EV i be connected at step t with $\tau = \tau_i(t)$ remaining actions and energy $E = E_i(t) \geq \hat{B}(\tau)$. Then $p^{\min}(E) \leq l_i(t) \leq u_i(t)$, and every commanded power $p \in [l_i(t), u_i(t)]$ leads to $E_i(t+1) \geq \hat{B}(\tau - 1)$.*

Proof. The bound $p^{\min} \le l_i$ holds because $p^{\min} \le 0 \le u_i$ and both arguments of the outer minimum in Equation 10 are at least $p^{\min}$. For $l_i \le u_i$ the outer minimum suffices, and the guard is inactive in exact arithmetic, because $g^{-1}\big((\hat{B}(\tau-1)-E)/\Delta t\big) \le \bar{P}(E)$ is equivalent to $\hat{B}(\tau-1) \le \varphi(E)$, and A1 with Equation 11 gives $\varphi(E) \ge \varphi(\hat{B}(\tau)) \ge \hat{B}(\tau-1)$. Turning to the second claim, let $y = (\hat{B}(\tau-1)-E)/\Delta t$, and since $l_i \ge \min\{u_i, g^{-1}(y)\} = g^{-1}(y)$ in exact arithmetic and g is increasing, every $p \ge l_i$ satisfies $g(p) \ge y$, and A3 gives $E_i(t+1) \ge E + \Delta t\, g(p) \ge \hat{B}(\tau-1)$. □

When $\tau \ge 2$, the EV is individually feasible at $t+1$ because $\tau_i(t+1) = \tau - 1$, and when $\tau = 1$, the lemma gives $E_i(d_i+1) \ge \hat{B}(0) \ge E_i^{\mathrm{req}}$.

S1.5 Feasibility preservation

The proof of Theorem 1 treats feasible steps, infeasible steps and the departure part in turn. Claim (a) uses only A3 and A6, and the departure part uses A1, A2 and A4 in addition. The conditions A1–A6 are those listed in Section 4.4, with A5 the definition of a feasible step.

Proof of Theorem 1. Claim (a) on feasible steps. The guard in Equation 10 gives $p_i^{\min} \le l_i \le u_i$ for every connected EV at every step, whether or not the EV is individually feasible, and empty chargers have $l_i = u_i = 0$. If the feasibility test of Algorithm S1 passes, Proposition 1(i) gives $\mathcal{K}(t) \ne \emptyset$, so $p(t) = \Pi_t(x(t)) \in \mathcal{K}(t)$ for every finite $x(t)$. The physical limits follow from $p_i(t) \in [l_i, u_i] \subseteq [p_i^{\min}(E_i(t)), \bar{P}_i(E_i(t))]$. For the transformer limit, $\mathbf{1}^\top p(t) \in [-C_{\mathrm{rev}}(t), \tilde{C}(t)]$, and A3 gives

$$\sum_i \tilde{p}_i(t) \in \Big[\sum_i p_i(t),\ \sum_i p_i(t) + n(t)\varepsilon_q\Big] \subseteq \big[-C_{\mathrm{rev}}(t),\ \tilde{C}(t) + n(t)\varepsilon_q\big] = \big[-C_{\mathrm{rev}}(t), C(t)\big].$$

Claim (a) on infeasible steps. Under A6, $\tilde{C}(t) = C(t) - n(t)\varepsilon_q \ge (N - n(t))\varepsilon_q \ge 0$ and $C_{\mathrm{rev}}(t) \ge 0$, so case (i) of the fallback rule needs $\tilde{C} < -C_{\mathrm{rev}} \le 0$ and is therefore impossible. Since $u_i \ge 0$, case (ii) with $\sum_i u_i < -C_{\mathrm{rev}} \le 0$ is impossible as well. Sub-case (iv) needs $b = \tilde{C} - \sum_i p_i^{\min} < 0$ and cannot occur because $p_i^{\min} \le 0$. Every infeasible step is therefore a step of case (iii), and Lemma S4 gives $p^{\min} \le p \le l \le u$ and $\mathbf{1}^\top p = \tilde{C} \in [-C_{\mathrm{rev}}, \tilde{C}]$. The physical limits follow, and A3 transfers the sum bound to the realised powers exactly as on feasible steps.

Claim (b). Fix an EV i with $p_i(t) \ge l_i(t)$ at every step of its stay, and consider the invariant $E_i(t) \ge \hat{B}_i(\tau_i(t))$. At arrival the invariant is A4. If it holds at step t, then $l_i(t)$ follows the first case of Equation 10, and every output of the projection or the fallback rule satisfies $p_i(t) \le u_i(t)$, so $p_i(t) \in [l_i(t), u_i(t)]$. Lemma S3 then gives $E_i(t+1) \ge \hat{B}_i(\tau_i(t) - 1)$. For $d_i > t$ this inequality is the invariant at $t+1$, and for $d_i = t$ it is the departure requirement $E_i(d_i+1) \ge \hat{B}_i(0) \ge E_i^{\mathrm{req}}$. The argument uses only the commands sent to EV i, so it holds whatever happens at the other chargers. Any EV whose stay contains no infeasible step receives $p_i \in [l_i, u_i]$ at every step of its stay and is covered. Since EV2Gym removes every EV by step $T-2$, every EV of the episode reaches its departure inside the horizon, and the departure part of the argument applies to all of them.

Claim (c). If A5 holds at every step, no step of the episode is infeasible, and claim (b) therefore applies to every EV of the episode. □

A2 enters through the remaining actions $\tau_i(t)$ that the bounds need at every step. Admission control never changes a requirement after arrival, so the requirement used in the invariant is fixed during the stay.

A policy-independent sufficient condition for A5. The lower bound l_i is non-increasing in E, because both terms of its first case are, and we let $l_i^{\max}(t)$ denote l_i evaluated at $E = \hat{B}_i(\tau_i(t))$. Suppose that, at every step, $C_{\mathrm{rev}}(t) \ge 0$, $\tilde{C}(t) \ge -C_{\mathrm{rev}}(t)$ and $\sum_{i \in \mathcal{N}(t)} l_i^{\max}(t) \le \tilde{C}(t)$. Under the invariant, $l_i(t) \le l_i^{\max}(t)$ and $\sum_i u_i \ge 0 \ge -C_{\mathrm{rev}}$, so A5 holds at t, and by induction A5 then holds for every policy. However, $l_i^{\max}(t) = \bar{P}_i(\hat{B}_i(\tau_i))$ whenever $\hat{B}_i(\tau_i) > E_i^{\min}$, so the condition asks the transformer to serve all non-trivially constrained EVs at full power at the same time.

S1.6 Fallback rule and deficit bound

When the feasibility test fails, the fallback rule replaces the projection. Since $l \le u$ holds by construction, three cases remain, the third with a sub-case.

(i) Contradictory limits, $\tilde{C} < -C_{\mathrm{rev}}$: the procedure of case (iii) is applied, so the forward limit has priority.

(ii) Excess PV export, $\sum_i u_i < -C_{\mathrm{rev}}$: every occupied charger receives $p_i = u_i$.

(iii) Excess required power, $\sum_i l_i > \tilde{C}$: every EV starts at $p_i^{\min}$, and the budget $b = \tilde{C} - \sum_i p_i^{\min}$ is distributed in increasing order of laxity, each EV receiving up to $l_i - p_i^{\min}$.

(iv) Sub-case of (i) and (iii) with $b < 0$: the output stays at $p = p^{\min}$, the closest admissible command to the forward limit.

In case (iii), ties are broken by the larger gap $l_i - p_i^{\min}$ and then by the lower charger index. The logs label the cases (i), (ii), (iii) and (iv) as FB-1*, FB-2, FB-1 and FB-0 with the numeric codes 3, 2, 1 and 0, and every analysis table uses this mapping. Under A6, cases (i), (ii) and (iv) cannot occur (proof of Theorem 1), so their counts in the result tables verify A6 on every replay.

Lemma S4 (Fallback)**.** *In case (iii) of the fallback rule with $b \ge 0$, the output satisfies $p^{\min} \le p \le l$ and $\mathbf{1}^\top p = \tilde{C}$, so the transformer limit and the physical limits hold. Every EV that is individually feasible at step t and receives $p_i = l_i$ stays individually feasible. In cases (ii) and (iv), the output minimises the transformer violation over the box $[p^{\min}, u]$, so no admissible command avoids that violation.*

Proof. In case (iii), $\sum_i (l_i - p_i^{\min}) > \tilde{C} - \sum_i p_i^{\min} = b$, so the budget is exhausted before every EV reaches l_i; hence $\mathbf{1}^\top p = \tilde{C}$ and $p \le l$. Outside case (i) we also have $\tilde{C} \ge -C_{\mathrm{rev}}$, and an individually feasible EV with $p_i = l_i$ stays feasible by Lemma S3. Over the box $[p^{\min}, u]$, the sum $\mathbf{1}^\top p$ is smallest at $p^{\min}$ and largest at u, so the outputs of cases (iv) and (ii) minimise the commanded violation. □

Lemma S5 (Deficit bound)**.** *Assume A1, A3 and A4, and assume that $\bar{P}_i$ is non-increasing in E. With $\eta^{\mathrm{c}} = \eta^{\mathrm{d}} = 1$ as in our EV2Gym instantiation, the total departure shortfall of the fleet in an episode is at most*

$$\Delta t \sum_{t \in \mathcal{T}_{\mathrm{iii}}} \Big(\sum_i l_i(t) - \tilde{C}(t) \Big) + \Delta t \sum_{t \in \mathcal{T}_{\mathrm{iv}}} \sum_i \big(l_i(t) - p_i^{\min}(t) \big), \tag{S1}$$

where $\mathcal{T}_{\mathrm{iii}}$ and $\mathcal{T}_{\mathrm{iv}}$ are the steps of cases (iii) and (iv).

Lemma S6 in Section S2 shows that EV2Gym's taper satisfies the extra condition of Lemma S5.

Proof. Let $\tilde{l}_i(t) = l_i(t)$ if $E_i(t) \ge \hat{B}_i(\tau_i(t))$ and $\tilde{l}_i(t) = u_i(t)$ otherwise, and define the feasibility deficit $\Delta E^{\mathrm{def}}(t) = (\hat{B}(\tau(t)) - E(t))^+$ with $\tau(d_i{+}1) = 0$. The shortfall $(E^{\mathrm{req}} - E(d_i{+}1))^+$ is at most $\Delta E^{\mathrm{def}}(d_i{+}1)$ because $\hat{B}(0) \ge E^{\mathrm{req}}$, and $\Delta E^{\mathrm{def}}(t_i^{\mathrm{arr}}) = 0$ holds by A4. It therefore suffices to show $\Delta E^{\mathrm{def}}(t+1) \le \Delta E^{\mathrm{def}}(t) + \Delta t \left(g(\tilde{l}) - g(p) \right)^+$ at every step and to sum these increments over the stay. If $E \ge \hat{B}(\tau)$, then $E + \Delta t\, g(l) \ge \hat{B}(\tau - 1)$ by Equation 10, and A3 gives $E(t+1) \ge \hat{B}(\tau - 1) - \Delta t\, (g(l) - g(p))^+$. When $E < \hat{B}(\tau)$ instead, $\tilde{l} = u$ and $E + \Delta t\, g(u) = \varphi(E)$, and Equation 11 with the 1-Lipschitz property of φ gives $\hat{B}(\tau - 1) - \varphi(E) \le \varphi(\hat{B}(\tau)) - \varphi(E) \le \hat{B}(\tau) - E = \Delta E^{\mathrm{def}}(t)$. Hence $\hat{B}(\tau - 1) - E(t+1) \le \Delta E^{\mathrm{def}}(t) + \Delta t\, (g(u) - g(p))$ in this case as well. For the fleet bound with g the identity, $p_i \ge \tilde{l}_i$ holds on feasible steps and $p = u$ holds in case (ii) of the fallback rule. In case (iii), $p \le l$ and $\sum_i (l_i - p_i) = \sum_i l_i - \tilde{C}$, and in case (iv), $p = p^{\min}$, so summing the per-step increments over all EVs gives Equation S1. □

S1.7 Conformal departure guarantee

Proof of Proposition 2. (i) By exchangeability, the rank of the test score $e_{n_{\mathrm{cal}}+1} = \hat{q}(\mathbf{w}_i) - d_i$ among the scores $e_1, \ldots, e_{n_{\mathrm{cal}}+1}$ is uniform up to ties, so $\mathbb{P}(e_{n_{\mathrm{cal}}+1} \le \hat{Q}_{1-\alpha}) \ge \lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / (n_{\mathrm{cal}}+1) \ge 1 - \alpha$ [2,3]. The event $e_{n_{\mathrm{cal}}+1} \le \hat{Q}_{1-\alpha}$ is equivalent to $d_i \ge \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha}$, and because d_i is an integer, also to $d_i \ge \lceil \hat{q}(\mathbf{w}_i) - \hat{Q}_{1-\alpha} \rceil$. Taking the maximum with t_i^{arr} changes nothing, because $d_i \ge t_i^{\mathrm{arr}}$ holds surely.

(ii) Consider the event that $d_i \geq \hat{d}_i^{\mathrm{lo}}$ and that $p_i(t) \geq l_i(t)$ at every step of the stay. On this event, Lemma S3 with τ^{lo}, applied up to $t = \hat{d}_i^{\mathrm{lo}}$, gives $E_i(\hat{d}_i^{\mathrm{lo}} + 1) \geq \hat{B}(0)$. For the remaining steps $\hat{d}_i^{\mathrm{lo}} < t \leq d_i$ we have $\tau^{\mathrm{lo}} = 1$, and the third property of Equation 11 gives $\hat{B}(1) \leq \max\{\hat{B}(0), E^{\min}\} \leq E_i(t)$. Lemma S3 with $\tau = 1$ then gives $E_i(t+1) \geq \hat{B}(0) \geq E^{\mathrm{req}}$ again, so $E_i(d_i + 1) \geq E^{\mathrm{req}}$ holds on the event, and the union bound gives the stated probability. If no step of the stay is infeasible almost surely, the second probability vanishes and the bound becomes $1 - \alpha$. The bound does not hold conditionally on the event that A5 held, because that event may be correlated with d_i. □

S2 EV2Gym modelling notes

This section documents how the model of the main text maps onto EV2Gym at commit `55ef368c`, and the simulator is used without changes to its source. All deviations from the defaults of the simulator are made through our own wrapper, configuration files and EV specification file.

S2.1 Charging taper and Assumption A1

In state-of-charge units $\varsigma = E/E^{\mathrm{cap}}$, let $\delta = \bar{P}^{\mathrm{nom}}\Delta t/E^{\mathrm{cap}}$ be the gain of one full-power step, $\varsigma^{\mathrm{tr}} \in (0, 1]$ the transition state of charge, m the taper multiplier and $\kappa = m\delta/(1 - \varsigma^{\mathrm{tr}})$. When the EV's maximum alternating-current charging power does not exceed the charger rating, reading the charging routine of EV2Gym at full current gives

$$\frac{\varphi(E)}{E^{\mathrm{cap}}} = \min\{\hat{\varphi}(\varsigma), \varsigma + \delta\}, \qquad \hat{\varphi}(\varsigma) = \begin{cases} \varsigma + \delta, & \varsigma + \delta \leq \varsigma^{\mathrm{tr}}, \\ 1 - (1 - \varsigma^{\mathrm{tr}}) \exp\Big(- m \dfrac{\varsigma + \delta - \varsigma^{\mathrm{tr}}}{1 - \varsigma^{\mathrm{tr}}} \Big), & \varsigma < \varsigma^{\mathrm{tr}} < \varsigma + \delta, \\ 1 - (1 - \varsigma)\, e^{-\kappa}, & \varsigma \geq \varsigma^{\mathrm{tr}}. \end{cases} \tag{S2}$$

According to a check script released with the code, the closed form agrees with EV2Gym's charging routine to 1.4×10^{-14} kWh. The check covers 768 combinations of EV model, efficiency, transition state of charge, multiplier $m \in \{1, 5\}$ and step length $\Delta t \in \{5, 15\}$ min. [TODO: The authors re-run and confirm this check].

Lemma S6 (A1 in EV2Gym)**.** *Suppose that the maximum alternating-current charging power of the EV does not exceed the charger rating. Then, for every step length, the full-power one-step map of EV2Gym in Equation S2 is continuous, non-decreasing and 1-Lipschitz.*

Proof. At $\varsigma = \varsigma^{\mathrm{tr}} - \delta$ the first two branches of $\hat{\varphi}$ both equal ς^{tr}, and at $\varsigma = \varsigma^{\mathrm{tr}}$ the last two branches both equal $1 - (1 - \varsigma^{\mathrm{tr}})e^{-\kappa}$. The map is therefore continuous, as a minimum of continuous functions. Its branch slopes are 1, $m \exp(\cdot) > 0$ and $e^{-\kappa} \in (0, 1)$, and $\varsigma + \delta$ has slope 1, so the map is non-decreasing as a minimum of non-decreasing functions, and A1 holds. Only the middle branch can have a slope above 1, namely when $m > 1$. On the interval $(\varsigma^{\mathrm{tr}} - \delta, \varsigma^{\mathrm{tr}})$, the function $\hat{h}(\varsigma) = \hat{\varphi}(\varsigma) - \varsigma - \delta$ is strictly concave with $\hat{h}(\varsigma^{\mathrm{tr}} - \delta) = 0$. If $\hat{h}(\varsigma_1) < 0$ at some interior point, concavity gives $\hat{h}'(\varsigma_1) \leq \hat{h}(\varsigma_1)/(\varsigma_1 - \varsigma^{\mathrm{tr}} + \delta) < 0$, so $\hat{\varphi}'(\varsigma_1) < 1$. Wherever the minimum in Equation S2 selects $\hat{\varphi}$ strictly, its slope is thus below 1, and elsewhere the minimum selects $\varsigma + \delta$ with slope 1. The map is therefore piecewise continuously differentiable with slopes in $[0, 1]$ and hence 1-Lipschitz. □

In EV2Gym, $\eta^{\mathrm{c}} = 1$ and $\varphi(E) = E + \Delta t\, \bar{P}(E)$, so a 1-Lipschitz φ is equivalent to a non-increasing $\bar{P}$, the extra condition of Lemma S5. A taper whose power drops faster than $1/(\eta^{\mathrm{c}}\Delta t)$ kW per kWh would violate A1, and certified thresholds would not repair the departure result.

S2.2 Actuation and Assumption A3

The decision variable p_i is the power that EV2Gym reports, bills and sends to the transformer, and it equals the battery-side energy change per step. With this convention $\eta_i^{\mathrm{c}} = \eta_i^{\mathrm{d}} = 1$ in Equation 1, and EV2Gym's charge and discharge efficiencies ζ_i^{c} and ζ_i^{d} act only as derating factors on the power limits.

Algorithm S1 One step of the safety layer

Require: raw action $x \in \mathbb{R}^N$ (non-finite entries set to 0); for each connected EV: $E_i(t)$, $\tau_i(t)$, certified table $\hat{B}_i$; limits $C(t)$, $C_{\mathrm{rev}}(t)$

for each occupied charger i **do**
  $u_i \leftarrow \bar{P}_i(E_i)$; $p_i^{\min} \leftarrow$ Equation 2
  if $E_i \geq \hat{B}_i(\tau_i) - 10^{-9}$ **then** $l_i \leftarrow \min\{u_i, \max\{p_i^{\min}, g_i^{-1}((\hat{B}_i(\tau_i - 1) - E_i)/\Delta t)\}\}$
  else $l_i \leftarrow u_i$
  end if
  $\chi_i \leftarrow (E_i - \hat{B}_i(\tau_i))/(\eta_i^{\mathrm{c}} \bar{P}_i^{\mathrm{nom}} \Delta t)$
end for
$\tilde{C} \leftarrow C - n(t)\varepsilon_q$
if $\sum_i l_i \leq \tilde{C}$ **and** $\sum_i u_i \geq -C_{\mathrm{rev}}$ **and** $\tilde{C} \geq -C_{\mathrm{rev}}$ **then** ▷ feasible step (A5)
  $p^0 \leftarrow \mathrm{clip}(x, l, u)$
  if $\mathbf{1}^\top p^0 \in [-C_{\mathrm{rev}}, \tilde{C}]$ **then** $p \leftarrow p^0$
  else
    sort breakpoints $\{x_i - u_i\} \cup \{x_i - l_i\}$; find λ^* by one pass
    $p \leftarrow \mathrm{clip}(x - \lambda^*, l, u)$; if $\mathbf{1}^\top p \notin [-C_{\mathrm{rev}}, \tilde{C}]$, recompute by safe bisection
  end if
else
  $p \leftarrow$ fallback rule of Section S1.6; log the case
end if
send p to the simulator: full current if $p_i \geq u_i - 10^{-3}$, else rounded upwards on the action grid
return p

Our specification file fixes one efficiency per EV model, 0.90 for seven models and 0.93 for one, taken from the simulator's table at the model's nominal current. EV2Gym rounds the normalised action, the command divided by $\zeta_i^{\mathrm{c}} P^{\mathrm{cs}}$ with the charger rating $P^{\mathrm{cs}} = 22.17$ kW, to five decimals and ignores discharge currents below 0.01 A. It also rounds the stored energy up to 0.01 kWh after every active step without billing the added energy. The actuation map therefore sends a command as full current when it lies within $\varepsilon_{\mathrm{sat}} = 10^{-3}$ kW of u_i, and otherwise rounds it upwards on the five-decimal action grid. Algorithm S1 lists one complete step of the safety layer with this actuation. Rounding can break the aggregate limit when $\sum_i l_i$ equals $\tilde{C}$ to machine precision. The implementation therefore checks the limit after each projection and recomputes a failing row by a bisection that keeps the feasible end of its bracket.

Lemma S7 (A3 in EV2Gym)**.** *Suppose that the maximum alternating-current charging power of every EV does not exceed the charger rating and that efficiencies are constant over current levels. Let chargers and EVs be three-phase, with one EV per charger and zero dead-bands for charging. Assume further that V2G is enabled and that all EVs leave before the episode ends, and let $z_0(m) \in (0, 1)$ be the positive root of $1 - e^{-mz} = z$ for $m > 1$. If $\kappa \leq \kappa^*(m) = m\, z_0(m)$ for every EV, the actuation map realises A3 with $\varepsilon_q = 0.008$ kW.*

Proof. We first show that every sub-maximal charging command $p \in (0, \bar{P}(E)]$, sent as the pilot $p/(\zeta^{\mathrm{c}} P^{\mathrm{cs}})$ of the normalised action, is delivered exactly. Let $\hat{\delta} = p\Delta t / E^{\mathrm{cap}}$ be its gain, with $\hat{\delta} \leq \hat{\delta}^* = \varphi(E)/E^{\mathrm{cap}} - \varsigma \leq \delta$. For a reduced pilot, EV2Gym shifts the transition point to $\varsigma_x = 1 - (\hat{\delta}/\delta)(1 - \varsigma^{\mathrm{tr}}) \geq \varsigma^{\mathrm{tr}}$, and we write $\Omega = 1 - \varsigma_x$. The delivered gain is $\min\{\hat{\delta}, \hat{\varphi}_x(\varsigma) - \varsigma\}$, where $\hat{\varphi}_x$ is the map of Equation S2 with $(\varsigma^{\mathrm{tr}}, \delta)$ replaced by $(\varsigma_x, \hat{\delta})$, and the exponent $m\hat{\delta}/\Omega$ of the shifted taper equals κ.

If $\varsigma + \hat{\delta} \leq \varsigma_x$, the step stays in the constant-current branch and the gain is $\hat{\delta}$. For $\varsigma \geq \varsigma_x$, the taper branch gives $\hat{\varphi}_x(\varsigma) - \varsigma = (1 - \varsigma)(1 - e^{-\kappa})$. Since $\varsigma \geq \varsigma^{\mathrm{tr}}$, Equation S2 gives $\hat{\delta} \leq \hat{\delta}^* \leq (1 - \varsigma)(1 - e^{-\kappa})$, and the gain is again $\hat{\delta}$.

In the remaining case $\varsigma < \varsigma_x < \varsigma + \hat{\delta}$, let $y = \varsigma + \hat{\delta} - \varsigma_x > 0$ and $\psi = y/\Omega$. Then $\hat{\varphi}_x(\varsigma) - \varsigma - \hat{\delta} = \Omega(1 - e^{-m\psi} - \psi)$, a concave function of ψ that vanishes at 0 and at $z_0(m)$, so the gain is $\hat{\delta}$ whenever $\psi \leq z_0(m)$. Since $\psi = 1 - (1 - \varsigma - \hat{\delta})/\Omega$, this condition reads $1 - \varsigma - \hat{\delta} \geq (1 - z_0)\, \hat{\delta}(1 - \varsigma^{\mathrm{tr}})/\delta$. If $\varsigma < \varsigma^{\mathrm{tr}}$, then $1 - \varsigma - \hat{\delta} \geq 1 - \varphi(E)/E^{\mathrm{cap}} \geq (1 - \varsigma^{\mathrm{tr}})e^{-\kappa}$ by Equation S2, and with $\hat{\delta} \leq \delta$ the condition holds if $e^{-\kappa} \geq 1 - z_0 = e^{-mz_0}$, that is, if $\kappa \leq mz_0$. For $\varsigma \geq \varsigma^{\mathrm{tr}}$, the bound $\hat{\delta} \leq (1 - \varsigma)(1 - e^{-\kappa})$ gives $1 - \varsigma - \hat{\delta} \geq (1 - \varsigma)e^{-\kappa}$. The right-hand side of the condition is then at most $(1 - z_0)(1 - \varsigma)(1 - e^{-\kappa})m/\kappa$,

so it suffices that $\kappa/(e^{\kappa}-1) \geq m(1-z_0)$. At $\kappa = mz_0$ the two sides are equal, because $e^{mz_0} = 1/(1-z_0)$, and the left side decreases in κ, so $\kappa \leq mz_0$ suffices again. Discharging has no taper, and EV2Gym delivers a discharge command within $[p^{\min}(E), 0)$ exactly, floored at $E^{\min}$ and capped by the discharge limit.

Next we bound the realised power. A saturated command delivers $\varphi(E) - E = \Delta t\, u_i$, and $u_i - p_i \leq \varepsilon_{\text{sat}} = 10^{-3}$ kW. Any positive command below $u_i - \varepsilon_{\text{sat}}$ is rounded upwards by at most $\zeta^{\text{c}} P^{\text{cs}} 10^{-5} \leq 2.22 \times 10^{-4}$ kW, stays below u_i and is delivered exactly by the first part, and EV2Gym's own rounding leaves a value on the grid unchanged. Negative commands are rounded towards zero by at most 2.22×10^{-4} kW, and if the dead-band applies, the realised power is 0 with $0 - p_i < 0.0069 + 2.22 \times 10^{-4}$ kW. Otherwise $\tilde{p}_i \geq p_i \geq p^{\min}(E)$, so neither the discharge limit nor the energy floor clips the command. In all cases $0 \leq \tilde{p}_i - p_i \leq 0.0072 < \varepsilon_q$ kW, and empty chargers draw zero power.

Finally, the stored energy is $E(t+1) = \lceil E + \Delta t\, \tilde{p}_i \rceil_{0.01} \geq E + \Delta t\, \tilde{p}_i \geq E + \Delta t\, p_i$. Also $E + \Delta t\, \tilde{p}_i \leq \varphi(E) \leq E^{\text{cap}}$, and since every capacity is a multiple of 0.01 kWh that is exactly representable after rounding, the upward rounding cannot exceed E^{cap}. □

With $m = 5$, the root is $z_0(5) = 0.99302$ and the threshold is $\kappa^*(5) = 4.965$. In our configuration, $\kappa \leq 2.391$ at $\Delta t = 15$ min and $\kappa \leq 0.797$ at $\Delta t = 5$ min for all eight EV models, so Lemma S7 applies. For $m \leq 1$ the delivery is not exact near the transition point, and the taper multiplier therefore keeps its default value of 5. Without the tightening of the forward limit by $n(t)\varepsilon_q$, realised powers would exceed the forward limit by up to about $n(t) \cdot 0.007$ kW on tight steps, according to a closed-loop check released with the code.

S2.3 Exogenous headroom (Assumption A6)

EV2Gym scales the peak of the inflexible load to $m_L P_{\text{tr}}$ with $m_L \sim \text{Normal}(0.4, 0.1^2)$ and clips the load at P_{tr}. PV generation is $\text{pv}(t)\, m_P P_{\text{tr}}$ with a normalised profile $\text{pv}(t) \leq 0.853 \cdot 1.1 < 0.94$ and $m_P \sim \text{Normal}(0.4, 0.1^2)$, both drawn once per episode. Hence $C(t) \geq P_{\text{tr}}(1 - m_L)$, and with $P_{\text{tr}} = \rho_{\text{tr}} N$ and $\rho_{\text{tr}} \geq 3$ kW, the condition $C(t) \geq N\varepsilon_q$ can fail only if $m_L > 1 - \varepsilon_q/\rho_{\text{tr}} \geq 0.997$, an event of about six standard deviations. Likewise $C_{\text{rev}}(t) \geq P_{\text{tr}}(1 - 0.94\, m_P)$ is negative only if $m_P > 1.06$, an event of more than six standard deviations. A6 therefore fails with a probability below 2×10^{-9} per episode. Since A6 is exogenous, any violation would show up as a step of case (i), (ii) or (iv) of the fallback rule, so the fallback counts of the result tables check it on every replay.

S2.4 Scenarios, arrival profiles and observation

As seen in Table S2, the grid quantities and the arrival rate scale linearly with the number of chargers N, whereas prices do not. The three arrival profiles of EV2Gym differ in their arrival times, stays and requested energies.

Workplace arrivals concentrate on weekday mornings, public arrivals spread over the day with an afternoon and evening peak, and private arrivals occur in the late afternoon and evening with overnight stays. EV2Gym does not generate an EV whose departure would fall within four steps of the episode end, so overnight sessions are silently missing when the episode starts too late. The public and private profiles therefore start at 12:00 instead of 05:00. On the scenario generator alone, the share of dropped spawn attempts then falls to 6.8% and 3.7%, and spawn multipliers per profile equalise the mean charger occupancy at about 0.21. They also equalise the energy demand per charger to within about 5%, and these scenario-generator statistics are design checks rather than results [TODO: the authors re-measure and confirm them]. The episode starts empty, and loads and PV are redrawn for every episode. Our literature configuration reproduces EV2Gym's 25-charger V2G profit-maximisation case with 100 kW, the workplace profile and no load or PV, using our EV specification file.

Table S3 lists the features of the two token types. All features are 0 for an empty charger, and empty chargers are also masked as attention keys. Forecasts are $\hat{P}(t+k) = \max\{0, P(t+k)(1 + 0.1\epsilon_{t,k})\}$ with independent standard normal $\epsilon_{t,k}$, and beyond the episode end the last true value is held. Every global feature is a ratio that does not grow with N, with $P_{\text{tr}}^{\text{nom}} = \max_t P_{\text{tr}}(t)$ and $N_{\text{day}} = 24/\Delta t$. The arrival profile is not a feature.

Table S2: Scenario parameters. Quantities marked "linear" scale with the number of chargers N.

Item	Setting	Scaling
Simulator	EV2Gym, commit `55ef368c`, unmodified; one EV per charger	–
Station sizes	train $N \in \{10, 20\}$; test $N \in \{5, 10, 20, 50, 100, 200\}$	–
Transformer rating	$P_{\mathrm{tr}} = \rho_{\mathrm{tr}} N$; $\rho_{\mathrm{tr}} \in \{4, 7\}$ kW (tight, moderate); stress level 3 kW at $N \in \{20, 200\}$	linear
Inflexible load	mean peak $0.4P_{\mathrm{tr}}$, clipped at P_{tr}, redrawn per episode	linear
PV generation	mean peak about $0.38P_{\mathrm{tr}}$, redrawn per episode	linear
Demand response	off	–
Step, horizon	$\Delta t = 15$ min, $T = 112$ (28 h)	–
Profiles (start, spawn multiplier, days)	workplace (05:00, 5, weekdays); public (12:00, 4, all days); private (12:00, 2.5, all days)	per charger
Mean occupancy	about 0.21 for all profiles	–
EV models	8 V2G-capable models; efficiency 0.90 (0.93 for one model); $E^{\min} = 5$ kWh; minimum stay 180 min	–
Taper multiplier	5 (EV2Gym default)	–
Desired energy	full battery	–
Prices	Dutch day-ahead prices; discharge price = charging price	–
Forecast noise	Gaussian, standard deviation 10% of the value	–
Test grid	36 configurations (N, ρ_{tr}, profile), 6 stress configurations and the literature configuration	–
Test episodes	100 per configuration for $N \le 50$ and the literature configuration, 50 for $N \ge 100$; 3,550 per policy	–

Table S3: Features of the charger token ($F_c = 12$) and the global token ($F_g = 38$) with their normalisation.

Charger token ξ_i	Global token ξ_0
1: occupancy flag	1–2: $c^{\mathrm{ch}}(t)$, $c^{\mathrm{dis}}(t)$ $/c^{\mathrm{ref}}$
2–3: $E_i(t)$, E_i^{cap} $/E^{\mathrm{ref}}$	3–10: $c^{\mathrm{ch}}(t+k)$, $k = 1, \dots, 8$, $/c^{\mathrm{ref}}$
4: state of charge E_i/E_i^{cap}	11–12: sine and cosine of the hour of day
5: remaining required energy $(E_i^{\mathrm{req}} - E_i)^+/E^{\mathrm{ref}}$	13: weekend flag; 14: remaining episode fraction $(T-t)/T$
6: remaining steps τ_i (conformal variant: τ_i^{lo}) $/N_{\mathrm{day}}$	15–16: $C(t)$, $C_{\mathrm{rev}}(t)$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$
7: laxity χ_i, clipped to $[-N_{\mathrm{day}}, 2N_{\mathrm{day}}]$, $/N_{\mathrm{day}}$	17–18: measured $P^{\mathrm{inf}}(t)$, $P^{\mathrm{pv}}(t)$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$
8–9: $\bar{P}_i^{\mathrm{nom}}$, P_i^{dis} $/P^{\mathrm{ref}}$	19–34: forecasts of P^{inf} and P^{pv} for $k = 1, \dots, 8$, $/P_{\mathrm{tr}}^{\mathrm{nom}}$
10: physical lower limit $p_i^{\min}(E_i)$ $/P^{\mathrm{ref}}$	35: occupancy ratio $n(t)/N$; 36: rating per charger $P_{\mathrm{tr}}^{\mathrm{nom}}/(NP^{\mathrm{ref}})$
11–12: bounds l_i, u_i $/P^{\mathrm{ref}}$	37–38: $\sum_i l_i$, $\sum_i u_i$ $/P_{\mathrm{tr}}^{\mathrm{nom}}$

S3 Request pipeline and perfect-information optima

S3.1 Station-level request pipeline

The arrival cap and admission control belong to the request setting of the station operator and not to the controller. For every test replay they are computed once, before any controller runs, and the resulting admitted requests E_i^{req} are shared by every compared method and by both optima. The wrapper writes the admitted requirement into EV2Gym's EV object at arrival, so every controller that reads EV2Gym's EV objects pursues the same target. For arrivals inside its horizon, the HiGHS eMPC uses the shared entries. Conformal passes re-certify the shared requirement with their bound $\hat{d}_i^{\mathrm{lo}}$, so their requirement is never larger, and passes without admission control keep the arrival-capped request. In every pass, shortfall is measured against the shared admitted requirement.

Arrivals are processed in time order, with ties broken by the charger index. An arriving EV j enters with its exogenous arrival energy, desired energy, departure step and EV parameters, and the cap of Equation 3 gives its first requirement with a certified threshold table. Admission control then solves the feasibility LP of Equation S3 at the arrival step t_0 over the EVs admitted earlier and still connected, together with EV j:

$$
\begin{aligned}
& E_i(t+1) = E_i(t) + \Delta t\, p_i(t), \qquad E_i^{\min} \le E_i(t+1) \le E_i^{\mathrm{cap}}, \qquad E_i(d_i+1) \ge E_i^{\mathrm{tgt}}, \\
& -P_i^{\mathrm{dis}} \le p_i(t) \le \bar{P}_i^{\mathrm{nom}}, \qquad p_i(t) \le \varrho_i^{\mathrm{in}}\,(E_i^{\mathrm{cap}} - E_i(t))\ (t > t_0), \qquad p_i(t_0) \in [p_i^{\min}, \bar{P}_i], \\
& -\hat{C}_{\mathrm{rev}}(t) \le \textstyle\sum_i p_i(t) \le \hat{C}(t) - n(t_0)\varepsilon_q.
\end{aligned}
\tag{S3}
$$

The energies of the earlier EVs at t_0 are read from a nominal plan and not from the trajectory of any

controller. After each admission decision, an elastic version of the same LP updates this plan so that the targets are reached as early as possible. The limits $\hat{C}$ and $\hat{C}_{\mathrm{rev}}$ use the measured values at t_0, the noisy forecasts for the next H_f steps and the last forecast afterwards. Its slope ϱ_i^{in} is the largest value with $\bar{P}_i(E) \geq \min\{\bar{P}_i^{\mathrm{nom}}, \varrho_i^{\mathrm{in}}(E_i^{\mathrm{cap}} - E)\}$ for all E, so the linear taper model is an inner approximation and every LP-feasible schedule is physically realisable.

Inside the LP, the new EV has the target $E_j^{\mathrm{tgt}} = E_j^{\mathrm{req}}$, and an earlier EV has the smaller of E_i^{req} and $\Psi_i^{\mathrm{LP}}(E_i(t_0)) - 10^{-6}$ as its target. Here Ψ_i^{LP} is the largest departure energy reachable in the LP's taper model, and this target is used only inside the admission LP. With full-battery requests as literal targets, the LP would be infeasible whenever such an EV is connected, because the linear taper model never reaches full charge in finitely many steps. Feasibility is monotone in the new request, so one LP that maximises $E_j(d_j + 1)$ gives the largest admissible request. Requests admitted earlier are never reduced, and the reduced request of EV j passes through the cap routine again, so A4 holds for every EV and every method. If the LP is infeasible even for the lowest request, the request is set to the arrival energy.

Each decision uses only information available at t_0, so a station operator can run the pipeline online. If its LP is feasible, an open-loop schedule exists that meets all targets from the nominal-plan energies under the forecast limits, provided that no further EV arrives. The pipeline does not guarantee A5 for any controller, because realised energies may lie below the nominal plan, forecasts are noisy and later arrivals add load. It therefore does not enter Theorem 1, and the evaluation passes without admission control measure its effect.

S3.2 Linear programs of the optima

Both perfect-information LPs know all sessions of the episode with their arrival and departure steps, arrival energies and admitted requirements, together with all prices and the realised limits $C(t)$ and $C_{\mathrm{rev}}(t)$. For every session i and step $t \in \mathcal{T}_i = \{t_i^{\mathrm{arr}}, \dots, d_i\}$ the variables are $p_i^+(t), p_i^-(t) \geq 0$ and $E_i(t)$, together with a shortfall slack $z_i \geq 0$ and, in the soft LP, overload slacks $o^{\pm}(t) \geq 0$. With the rounding allowance $G_i^{\max} = 0.01\,(d_i - t_i^{\mathrm{arr}} + 1)$ kWh, the constraints are

$$
\begin{aligned}
&E_i(t_i^{\mathrm{arr}}) = E_i^{\mathrm{arr}}, \qquad E_i(t+1) = E_i(t) + \Delta t\,\big(p_i^+(t) - p_i^-(t)\big), \qquad E_i^{\min} - G_i^{\max} \leq E_i(t+1) \leq E_i^{\mathrm{cap}},\\
&p_i^+(t) \leq \bar{P}_i^{\mathrm{nom}}, \qquad p_i^-(t) \leq P_i^{\mathrm{dis}}, \qquad p_i^+(t_i^{\mathrm{arr}}) \leq \bar{P}_i(E_i^{\mathrm{arr}}), \qquad p_i^+(t) \leq \varrho_i^{\mathrm{out}}\big(E_i^{\mathrm{cap}} - E_i(t)\big)\ (t > t_i^{\mathrm{arr}}),\\
&E_i(d_i+1) + z_i \geq E_i^{\mathrm{req}} - G_i^{\max}, \qquad -C_{\mathrm{rev}}(t) - o^-(t) \leq \textstyle\sum_i \big(p_i^+(t) - p_i^-(t)\big) \leq C(t) + o^+(t),
\end{aligned}
\tag{S4}
$$

where the hard LP fixes $o^{\pm} = 0$. The profit of a schedule is $\Pr(p) = \sum_t \Delta t \sum_i \big(c^{\mathrm{dis}} p_i^- - c^{\mathrm{ch}} p_i^+ - c^{\mathrm{thr}}(p_i^+ + p_i^-)\big)$, with the throughput cost c^{thr} equal to zero in all reported runs.

Lemma S8 (Taper rows)**.** *Let every EV satisfy the conditions of Lemma S7, and let* $\varrho_{\kappa,i} = (1 - e^{-\kappa_i})/\Delta t$. *Then* $\bar{P}_i(E) \leq \min\{\bar{P}_i^{\mathrm{nom}}, \varrho_{\kappa,i}(E_i^{\mathrm{cap}} - E)\}$ *for every* $E \in [E^{\min}, E^{\mathrm{cap}}]$. *Equality holds except on the one-step transition band* $\varsigma \in (\varsigma_i^{\mathrm{tr}} - \delta_i, \varsigma_i^{\mathrm{tr}})$, *and* $\sup_E \bar{P}_i(E)/(E_i^{\mathrm{cap}} - E) = \varrho_{\kappa,i}$.

Proof. In state-of-charge units, $\bar{P}\Delta t/E^{\mathrm{cap}} = \varphi/E^{\mathrm{cap}} - \varsigma$ by Equation S2, and for $\varsigma \geq \varsigma^{\mathrm{tr}}$ this gain is $\min\{\delta, (1-\varsigma)(1-e^{-\kappa})\}$ and equals the right-hand side. The constant-current region $\varsigma + \delta \leq \varsigma^{\mathrm{tr}}$ has the gain δ, and the right-hand side equals δ if and only if $\delta \leq (1-\varsigma)(1-e^{-\kappa})$. Since $1 - \varsigma \geq 1 - \varsigma^{\mathrm{tr}} = m\delta/\kappa$, it suffices that $\kappa \leq m(1 - e^{-\kappa})$, and with $\kappa = mz$ this is $z \leq 1 - e^{-mz}$, that is, $\kappa \leq \kappa^*(m)$. On the transition band, let $y = \varsigma^{\mathrm{tr}} - \varsigma \in (0, \delta)$, so that $1 - \varphi/E^{\mathrm{cap}} \geq (1 - \varsigma^{\mathrm{tr}})e^{-m(\delta - y)/(1-\varsigma^{\mathrm{tr}})}$. With $e^v \geq 1 + v$ and $m \geq 1$, the right-hand side of this bound is at least $e^{-\kappa}(1 - \varsigma^{\mathrm{tr}} + my) \geq e^{-\kappa}(1 - \varsigma)$. The gain is therefore at most $(1-\varsigma)(1-e^{-\kappa})$, and it is at most δ anyway, so the inequality holds everywhere, and the supremum follows from equality in the upper region. □

The code uses $\varrho_i^{\mathrm{out}} = 1.01\,\varrho_{\kappa,i}$, a valid outer row that is 1% looser than exact. A check script released with the code confirms the inequality to 1.7×10^{-13} kW and the equality off the band on 320 configurations. With these rows the LPs model EV2Gym's charging limit up to the step discretisation of the transition band and the 1% margin. The remaining relaxations are the rounding allowance and simultaneous charging and discharging, and the latter is cost- and energy-neutral, and hence exact, when $\eta = 1$, $c^{\mathrm{dis}} = c^{\mathrm{ch}}$ and $c^{\mathrm{thr}} = 0$, as in all reported runs.

S3.3 Upper bounds

The hard LP is solved lexicographically with HiGHS [4]. Stage 1 computes the least total shortfall $z^* = \min \sum_i z_i$ with $G^{\max} = 0$, and stage 2 maximises $\Pr(p)$ with the allowance and $\sum_i z_i \le z^* + 10^{-6}$, giving $J^{\mathrm{LP,ceil}}$. The same stage 2 with $G^{\max} = 0$ gives J^{LP}, reported only in Section S5. In contrast, the soft LP maximises $\Pr(p) - w_E \sum_i z_i - w_P \Delta t \sum_t (o^+(t) + o^-(t))$ subject to Equation S4 in a single stage, giving $J^{\mathrm{pen,LP}}$.

Proposition S2 (Hard optimum)**.** *Let the configuration of Lemma S7 hold, and let π be any controller whose reported powers satisfy Equation 5 at every step and whose realised total shortfall against the admitted requirements is at most z^*. Then $J^\pi \le J^{\mathrm{LP,ceil}}$.*

Proof. Let $\tilde{p}_i(t)$ be the reported powers and $E_i^{\mathrm{nom}}(t) = E_i^{\mathrm{arr}} + \Delta t \sum_{s<t} \tilde{p}_i(s)$ the energy that the billed powers alone would produce. EV2Gym bills exactly $\tilde{p}$, so J^π equals Pr at $p^\pm = \tilde{p}^\pm$. The rounding adds less than 0.01 kWh per active step, so $E_i^{\mathrm{real}}(t) - G_i^{\max} \le E_i^{\mathrm{nom}}(t) \le E_i^{\mathrm{real}}(t) \le E_i^{\mathrm{cap}}$ along the trajectory. Both the lowered energy floor and the lowered departure row therefore hold for E^{nom} with $z_i = (E_i^{\mathrm{req}} - E_i^{\mathrm{real}}(d_i + 1))^+$, and the power rows hold because $\tilde{p}_i \le \bar{P}_i(E_i^{\mathrm{real}}) \le \bar{P}_i^{\mathrm{nom}}$ and $\tilde{p}_i \ge -P_i^{\mathrm{dis}}$. Moreover, the first-step row holds because $E^{\mathrm{nom}}(t_i^{\mathrm{arr}}) = E_i^{\mathrm{arr}}$. For the taper rows, EV2Gym never delivers more than $\varphi(E) - E$, so Lemma S8 gives $\tilde{p}_i(t) \le \varrho_{\kappa,i}(E^{\mathrm{cap}} - E_i^{\mathrm{real}}(t)) \le \varrho_i^{\mathrm{out}}(E^{\mathrm{cap}} - E_i^{\mathrm{nom}}(t))$. The transformer rows are the assumption, and $\sum_i z_i \le z^*$, so the trajectory is feasible for stage 2 and its objective equals J^π. □

Proposition S3 (Soft-penalty optimum)**.** *Let the configuration of Lemma S7 hold, and let commands be clipped to the physical limits of Equation 2. For every simulated trajectory on a replay, with shortfall measured against the shared admitted requirement, $J^{\mathrm{pen}}(\pi) \le J^{\mathrm{pen,LP}}$.*

Proof. Take $p^\pm = \tilde{p}^\pm$, $E = E^{\mathrm{nom}}$, $z_i = \Delta E_i^{\mathrm{sh}}$ and $o^\pm(t)$ equal to the realised overloads in either direction. The energy, power, first-step and taper rows hold as in the proof of Proposition S2, because they use only the physics of EV2Gym and not any constraint that the controller respects. For the departure row, $E^{\mathrm{nom}}(d_i + 1) + z_i \ge E^{\mathrm{real}}(d_i + 1) - G_i^{\max} + z_i \ge E_i^{\mathrm{req}} - G_i^{\max}$, and the transformer rows hold by the definition of $o^\pm$. At this point the objective equals $J^{\mathrm{pen}}(\pi)$, with the same billed powers, shortfall and overload energy. □

The bound is a property of the measurement, not of the controller, so it covers controllers with shortfall or overload, the conformal and no-admission passes and the rule-based and model-predictive baselines. Every pass in the wrapper applies the physical clip. In the reported runs, none of the 153,100 episodes exceeded the soft-penalty optimum by more than 10^{-6} € (analysis load report). A unit test released with the code checks $J^{\mathrm{pen}} \le J^{\mathrm{pen,LP}}$ for all rule-based methods on small replays, including random actions. Inside EV2Gym, a controller can exceed J^{LP} through the unbilled rounding gain, so the main text uses $J^{\mathrm{LP,ceil}}$ for the cost of the guarantee and reports J^{LP} only here.

S4 Compared methods, training and statistics

S4.1 Implementation of the compared methods

Every method and every evaluation pass receives the shared admitted requirement of Section S3, and shortfall is always measured against it. Methods without the safety layer execute only the physical clip $\mathrm{clip}(x_i, p_i^{\min}, u_i)$, and variants with the layer pass their output as the raw action x through Algorithm S1, so they inherit Theorem 1.

Penalty-based methods. Attention+penalty, MLP+penalty and the variant without departure bound learn from the reward $r(t)$ minus two penalties. They are w times the departure shortfall of the EVs that leave at step t and w times the transformer overload energy of the step. Shortfall is measured against the admitted requirement, overload uses the untightened limits and the reported powers, and the weight $w \in \{1, 10, 100\}$ €/kWh is tuned. At the evaluation weight of 10 €/kWh, more than eleven times the highest hourly price, no violation pays at the optimum of the penalised objective, although learned policies can still violate.

Attention+penalty and MLP+penalty use $\beta = 0$, and their actor loss masks only tokens whose action cannot influence the commanded power. The variant without departure bound runs the safety layer with $l_i = p_i^{\min}$, so it keeps $\beta = 0.01$ and masks fallback steps like SA-Attn.

Attention+PPO-Lagrangian. This baseline solves the constrained problem $\max J_r$ subject to expected episode shortfall and overload energy of at most zero [5,6]. The attention policy runs with the physical clip and the profit reward, and the critic has three heads, one for the value and one for each cost, each with its own advantage estimate. PPO uses the combined advantage $(A_r - \Lambda_E A_E - \Lambda_P A_P)/(1 + \Lambda_E + \Lambda_P)$, normalised per minibatch. Once per rollout, $\Lambda_k \leftarrow \mathrm{clip}(\Lambda_k + \eta_\Lambda(\hat{J}_k - 0), 0, 1000)$ for $k \in \{E, P\}$, with $\Lambda_k = 0$ at the start and the step size $\eta_\Lambda \in \{0.5, 2, 8\}$ tuned. The cost limit of zero matches the hard requirements of the problem, and the cap of 1000 keeps the multipliers finite while violations persist, and the training logs record their trajectories.

Size-agnostic and fixed-size networks. Deep Sets+safety layer embeds each charger token with a shared network and adds the masked mean of all embeddings and the global token as context. It decodes one Gaussian per token, so no token attends to another. The MLP baselines map the zero-padded concatenation of 20 charger tokens and the global token through two hidden layers of 640 units. Their width matches the parameter count of the attention actor (614,440 versus 610,370), and the MLP baselines are evaluated and reported only at $N \leq 20$.

Rules. The rules as fast as possible (AFAP), as late as possible (ALAP), round robin and LLF are price-unaware and charge only. For each EV, sLLF [7] computes the next-step laxity $\chi_i^0 = (E_i - \hat{B}_i(\tau_i - 1))/(\bar{P}_i^{\mathrm{nom}} \Delta t)$ and the rate $p_i = \mathrm{clip}(\bar{P}_i^{\mathrm{nom}}(L - \chi_i^0), 0, u_i)$, with a common level L found by bisection so that $\sum_i p_i = \max(0, \tilde{C})$. It is the published rule with the peak rate $\bar{P}_i^{\mathrm{nom}}$ and a cap that includes the taper. The price-threshold rule compares the current price with the median of all prices seen so far together with the known price window. Below the median it allocates $\tilde{C}(t)$ by LLF, above the median it requests the maximum V2G discharge, and otherwise it idles.

Model-predictive baselines. eMPC-HiGHS is our re-implementation of the eMPC of EV2Gym [8] as a receding-horizon net-power LP with HiGHS. Its LP uses per-EV limits, exact first-step bounds including the taper and the admitted requirements with an elastic departure constraint of 1000 €/kWh. Prices and arrivals are known exactly within its horizon of 10 steps, whereas load and PV come from EV2Gym's noisy forecasts. A 24-h horizon variant and the original Gurobi-based eMPC were planned but not run.

External learned baselines. The heterogeneous graph policy (EV-GNN) [9] was planned as an external size-agnostic baseline, to be run from its public code at commit `01aa1b7a` with its own training script, but this study did not include it. In addition, the graph-enhanced decision transformer (GNN-DT) [10] has public code under an open-source licence (DT4EVs, commit `8b327d12`). GNN-DT trains offline on trajectories that an optimiser generates for each scenario inside a modified simulator copy, and porting this pipeline to our scenarios was not done.

S4.2 Training loop

Algorithm S2 lists the training loop of SA-Attn, in which the safety layer of Algorithm S1 acts as part of the environment. Each environment writes into its own buffer, and at update time all token tensors are right-padded to the largest N in the batch with masked entries. Padding changes neither the actor outputs of real tokens nor the value (Lemma S1), and the log-probability, entropy and regulariser sums run over masks only. The learning reward is the profit divided by Nr^{ref}, so that returns of stations with different N have the same scale, and every reported result uses the undivided profit.

S4.3 Hyperparameters and tuning

Table S4 lists the hyperparameters. Each of the seven tuned learned methods (SA-Attn, Attention+penalty, Attention+PPO-Lagrangian, Attention+safety layer without departure bound, Deep Sets+safety layer, MLP+safety layer and MLP+penalty) received 12 configurations. They consist of the default point and 11 points sampled without replacement from the grid of Table S4 with a fixed sampling seed per method. Each configuration is trained on the training mixture for 0.5×10^6 steps with seed 0 and evaluated on 24 validation episodes that cycle through the training configurations, with seeds disjoint from the test replays. Selection uses the mean validation J^{pen} at $w = 10$ €/kWh for every

Algorithm S2 PPO training of SA-Attn with a variable number of chargers

Require: 16 environments (8 with $N = 10$, 8 with $N = 20$); actor π_θ and critic V_ω

while fewer than 3×10^6 environment steps **do**
  for 128 steps in every environment **do**
    observe tokens, l, u, $\tilde{C}$, C_{rev} and the feasibility flag
    sample $a \sim \pi_\theta(\cdot \mid s)$; $x \leftarrow P^{\text{ref}} a$
    $p \leftarrow \Pi_t(x)$ if feasible, else fallback rule ▷ Algorithm S1
    step the simulator; $r \leftarrow$ realised profit of the step (Equation 6); store $r/(Nr^{\text{ref}})$, a, $\log \pi_\theta(a \mid s)$ over tokens with $u_i - l_i > 10^{-9}$
  end for
  compute advantages with GAE ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$); pad tokens to the largest N in the batch
  for 10 epochs, minibatches of n_{mb} transitions **do**
    clipped surrogate ($\epsilon_{\text{clip}} = 0.2$) on feasible transitions with normalised advantages, divided by n_{mb}
    loss $\leftarrow -$surrogate $+ c_{\text{vf}}\cdot$value loss $- c_{\text{ent}}\cdot$entropy $+ \mathcal{L}_{\text{reg}}$ (Equation 13, feasible steps only)
    Adam step with gradient-norm clipping 0.5; stop the epochs early if the approximate KL divergence exceeds 0.03
  end for
end while

method. The conformal variant and the reference trained at $N = 50$ inherit the selection of SA-Attn, and β and the architecture are fixed for all methods with the safety layer. Tuning ran 84 training and 84 validation jobs, and the selected configurations are released with the code.

EV2Gym could not build some training episodes, because negative residential loads gave a negative scale of the forecast noise. The training seeds were therefore redrawn, once per tuning job and twice per training job, and the test replays were built unchanged.

S4.4 Statistical analysis

Each learned method was trained with two seeds, and the reference trained at $N = 50$ also with two. With two seeds, a seed-level t-test has one degree of freedom, so no seed-level test or interval is computed. For each test replay, the value of a learned method is the mean over its two seeds, the seed-averaged policy, and all methods run on the same fixed replays. Tables report the mean over configurations of the replay means of the seed-averaged policy, followed by the range of the same statistic over the two seeds. Deterministic rules and eMPC-HiGHS run once and carry no range. Confidence intervals come from a percentile bootstrap with 10^4 resamples of the test replays within each configuration, and the interquartile mean (IQM) of the per-episode gap uses 2×10^3 resamples [11].

Comparisons with SA-Attn run per station size on the metrics of each results table, and each safety-layer method is also compared with its penalty-based twin. The primary test is a two-sided Wilcoxon signed-rank test [12] on the replay-level differences of the seed-averaged policies, with the matched-pairs rank-biserial correlation as effect size. Holm's correction runs over all comparisons of one test file, and each file covers one results table, with up to 351 comparisons per family. The Wilcoxon test concerns the median shift of the differences, whereas the bootstrap interval concerns their mean. A few episodes with large penalties can move the mean without moving the median, so the two can disagree, and such comparisons are reported as inconclusive. Between-seed variability is not captured by any test, and results over few seeds can mislead [13], so the seed ranges and the seed count are reported as a limitation.

S4.5 Compute

The runs used an AMD Ryzen 7 7700 with 8 cores and 16 threads and an NVIDIA GeForce RTX 5070 Ti with 16 GB. WSL2 was assigned 12 threads and 31 GB of memory. As software, Windows 11 Pro with WSL2 Ubuntu 24.04, Python 3.13.15 and PyTorch 2.14.1 with CUDA 13.0 were used. Preparing scenarios, admission tables and conformal calibrations took 1.6 h, tuning 7.9 h and training of 18 runs 10.2 h. Evaluation with 298 evaluation and 43 optimum jobs took 2.5 h, and the analysis 6 min. The

Table S4: Hyperparameters and tuning grid.

Group	Parameter	Value
Network	width d_m, layers n_L, heads n_H	128, 3 (pre-LayerNorm), 4
	feed-forward width, dropout	512 (GELU), 0
	actor head (shared), critic readout	$128 \to 64 \to 2$; attention pooling with one query, $128 \to 64 \to 1$
	$\log \sigma$ clamp and initial value	$[-5, 0.5]$, -0.5
	parameters (actor / critic)	610,370 / 676,481; small network 228,067 in total (head width 32)
	precision	float32 networks, float64 safety layer
PPO	optimiser	Adam, $\epsilon = 10^{-5}$, linear learning-rate decay
	learning rate (tuned)	$\{1, 3, 10\} \times 10^{-4}$
	minibatch n_{mb} (tuned)	$\{256, 512, 1024\}$
	entropy coefficient c_{ent} (tuned)	$\{0, 10^{-3}, 10^{-2}\}$
	discount γ, GAE λ_{GAE}	0.99, 0.95
	clip range ϵ_{clip}, value coefficient c_{vf}	0.2, 0.5
	environments, rollout length	16 (8 with $N = 10$, 8 with $N = 20$), 128
	epochs, target KL, gradient-norm clip	10, 0.03, 0.5
	advantage normalisation	per minibatch
	steps per run, seeds	3×10^6; 2 (seeds 0 and 1)
Penalty reward	training weight w (tuned), β	$\{1, 10, 100\}$ €/kWh; $\beta = 0$, except 0.01 without departure bound
Lagrangian	dual step η_Λ (tuned), cost limit, $\Lambda_{\max}$	$\{0.5, 2, 8\}$, 0, 1000
Safety layer	P^{ref}, β	11 kW, 0.01
	ε_q, $\varepsilon_{\mathrm{sat}}$, tolerance	0.008 kW, 0.001 kW, 10^{-9} kW or kWh
Features	E^{ref}, c^{ref}, H_f	100 kWh, 0.25 €/kWh, 8 steps
Evaluation	weights w_E, w_P in J^{pen}	10 €/kWh each, all methods
	test action	deterministic, $a = \mu_\theta$
Conformal	n_{cal}, α	5,000 per configuration, $\{0.01, 0.05, 0.1, 0.2\}$
	quantile predictor	gradient boosting, 200 trees of depth 3, 2,000 training sessions

core set produced 153,100 episodes of 30 methods, and no job failed. Table S13 gives the training time per run and the inference time per control step. Without retraining, the conformal variant is evaluated at all four values of α, because Proposition 2 holds for any policy.

S5 Additional results

[TODO: The authors check every reported value against the analysis outputs and revise the wording of this section].

The tables of this section are generated by the analysis pipeline. Its files T2, T3, T3b, T3c, T3d, T5, T4, T7, T6, TS1, TS2 and TS3 appear as Tables S5 to S16 in this order. Values are means of the seed-averaged policies with the range over the two training seeds in parentheses, and † marks deterministic methods with a single run. The example-day figure, the planned ablations and sensitivity studies other than the stress rating were not run.

S5.1 Learning curves and zero-shot transfer

Figure S1 shows the training curves of the learned methods, with bands over the two seeds. Table S5 lists the zero-shot results behind Figure 3 for every station size, including the transfer gap to the reference trained at $N = 50$. That reference is in distribution only at $N = 50$, so at $N = 100$ and $N = 200$ the transfer gap compares two extrapolations.

S5.2 Constraint satisfaction

Pooled over the test grid, Table S6 gives E4, energy not served, overload, the shortfall relative to the deficit bound of Equation S1 and the capped and admission-reduced requests. The deficit-bound ratio covers only the methods with the full safety layer, because the variant without departure bound has no

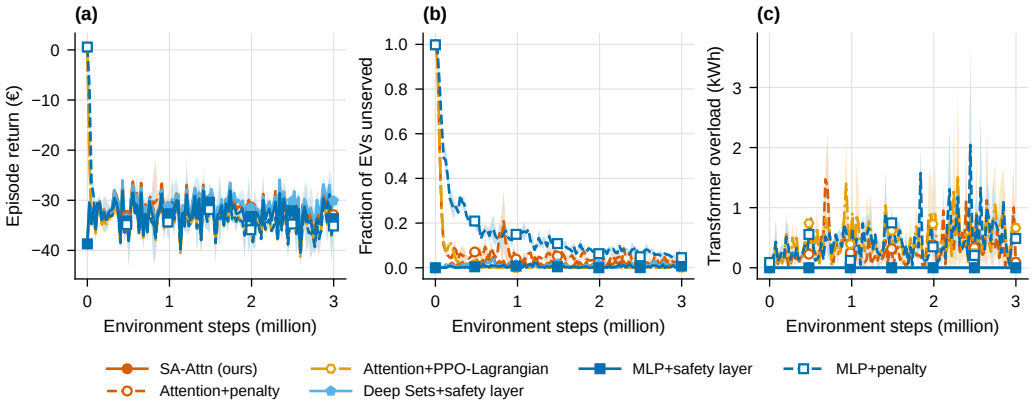


Figure S1: Training curves of the learned methods versus environment steps; bands show the range over the two training seeds.

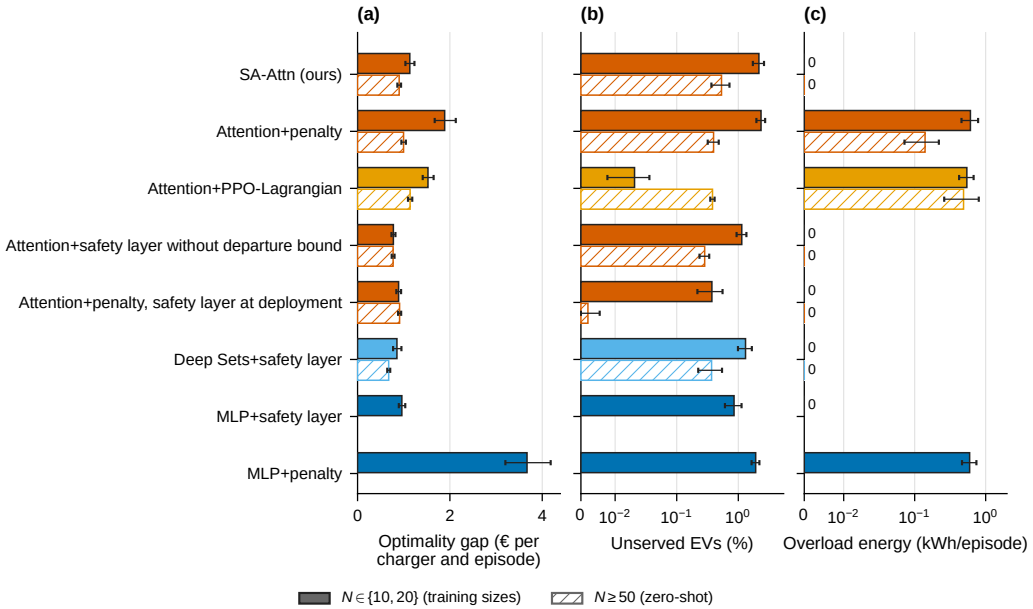


Figure S2: Gap of the architecture and constraint-handling variants of Table S11 at the training sizes and in zero-shot transfer.

such bound. Per station size, Table S7 gives E4, the infeasible steps by fallback case and the binding rates of the aggregate limits and of the departure lower bound. Table S8 and Table S9 give event counts summed over all episodes of both seeds, overall and per station size. Their last column is the episode-level check of Theorem 1(b): unserved EVs in episodes without any infeasible step. It is zero for every non-conformal method with the departure bound, whereas the variant without departure bound had 1,551 such EVs. Over all episodes, however, that variant left fewer EVs unserved than SA-Attn, 1,551 against 6,158 (Section 6.4). For the conformal variants, the check also counts EVs whose re-certified requirement lay below the shared one or that left before the predicted bound, cases that the theorem does not cover.

S5.3 Cost of the guarantee and constraint handling

Table S10 gives the cost of the guarantee behind Figure 5, in J^{pen} and in profit, for the attention and MLP pairs. Figure S2 and Table S11 compare the architectures and constraint-handling variants at the training sizes and in zero-shot transfer. At $N \geq 50$, SA-Attn had a gap of 0.901, Deep Sets+safety layer 0.674 and Attention+safety layer without departure bound 0.770. The penalty-trained policy with the layer at deployment reached 0.910, Attention+penalty 0.996 and Attention+PPO-Lagrangian 1.138.

Table S5: Zero-shot transfer across station sizes N for policies trained at N ∈ {10, 20} (profiles and $\rho_{\mathrm{tr}} \in \{4, 7\}$ kW pooled; per episode): learned methods, SA-Attn trained at N = 50 (reference) and the three baselines with the safety layer that have the lowest mean optimality gap.

Method	Optimality gap (€ per charger and episode)	J^{pen} (€/charger)	Profit (€/charger)	Unserved EVs (%)	Overload (kWh)	Transfer gap to the N = 50 reference (€/charger) [95 % CI]
N = 5						
SA-Attn (ours)	2.292 (2.080–2.503)	−3.49	−2.00	7.50	0	–
Deep Sets+safety layer	1.663 (1.586–1.739)	−2.86	−1.84	5.01	0	–
Attention+safety layer without departure bound	1.026 (0.947–1.105)	−2.23	−1.90	2.80	0	–
Attention+penalty, safety layer at deployment	1.049 (0.976–1.121)	−2.25	−2.00	2.00	0	–
Attention+penalty	4.561 (4.014–5.109)	−5.76	−1.91	5.28	1.04	–
Attention+ PPO-Lagrangian	3.742 (2.731–4.752)	−4.94	−2.23	0	1.36	–
MLP+safety layer	1.895 (1.741–2.049)	−3.10	−1.98	5.73	0	–
MLP+penalty	4.730 (4.715–4.746)	−5.93	−2.24	0.09	1.80	–
Price-threshold+ safety layer	0.956†	−2.16†	−1.93†	1.91†	0†	–
ALAP+safety layer	1.107†	−2.31†	−2.17†	1.72†	0†	–
AFAP+safety layer	1.038†	−2.24†	−2.24†	0†	0†	–
N = 10 (training size)						
SA-Attn (ours)	1.290 (1.255–1.325)	−2.40	−1.94	3.15	0	–
Deep Sets+safety layer	0.984 (0.940–1.028)	−2.10	−1.77	1.92	0	–
Attention+safety layer without departure bound	0.776 (0.733–0.819)	−1.89	−1.82	1.33	0	–
Attention+penalty, safety layer at deployment	0.908 (0.885–0.930)	−2.02	−1.94	0.63	0	–
Attention+penalty	2.204 (2.172–2.237)	−3.32	−1.90	2.75	0.74	–
Attention+ PPO-Lagrangian	1.792 (1.330–2.255)	−2.91	−2.19	0	0.72	–
MLP+safety layer	1.025 (1.012–1.037)	−2.14	−1.93	1.69	0	–
MLP+penalty	5.547 (2.529–8.564)	−6.66	−2.18	2.84	0.75	–
Price-threshold+ safety layer	0.790†	−1.90†	−1.89†	0.33†	0†	–
ALAP+safety layer	1.007†	−2.12†	−2.12†	0.21†	0†	–
AFAP+safety layer	1.095†	−2.21†	−2.21†	0†	0†	–
N = 20 (training size)						
SA-Attn (ours)	0.973 (0.971–0.976)	−2.26	−2.16	1.17	0	–
Deep Sets+safety layer	0.722 (0.687–0.758)	−2.01	−1.96	0.71	0	–
Attention+safety layer without departure bound	0.776 (0.709–0.843)	−2.06	−2.03	0.95	0	–
Attention+penalty, safety layer at deployment	0.871 (0.867–0.874)	−2.16	−2.15	0.11	0	–
Attention+penalty	1.569 (1.315–1.823)	−2.86	−2.12	1.92	0.48	–
Attention+ PPO-Lagrangian	1.258 (1.076–1.440)	−2.55	−2.36	0.04	0.37	–
MLP+safety layer	0.894 (0.875–0.913)	−2.18	−2.18	0.02	0	–
MLP+penalty	1.795 (1.795–1.795)	−3.08	−2.38	0.99	0.44	–
Price-threshold+ safety layer	0.788†	−2.08†	−2.06†	0.28†	0†	–
ALAP+safety layer	1.028†	−2.32†	−2.32†	0†	0†	–
AFAP+safety layer	1.099†	−2.39†	−2.39†	0†	0†	–
N = 50						
SA-Attn (ours)	0.921 (0.886–0.956)	−2.15	−2.10	0.64	0	0.037 [0.026, 0.050]
Deep Sets+safety layer	0.698 (0.675–0.720)	−1.93	−1.89	0.48	0	−0.186 [−0.204, −0.170]
SA-Attn trained at N = 50 (reference)	0.884 (0.833–0.935)	−2.12	−2.07	0.57	0	–

S5.4 Conformal departure bound

Figure S3 and Table S12 show coverage, shortfall against the shared requirement, gap and charged energy of SA-Attn-CP against the miscoverage level α. The achieved coverage was 0.992, 0.950, 0.902 and 0.799 for the targets 0.99, 0.95, 0.90 and 0.80. Summed over all episodes, the shortfall was 771,853 kWh at $\alpha = 0.01$, 12,507 kWh at $\alpha = 0.05$ and 2,568 kWh with known departure steps. For comparison, the

Table S5: (continued)

Method	Optimality gap (€ per charger and episode)	J^{pen} (€/charger)	Profit (€/charger)	Unserved EVs (%)	Overload (kWh)	Transfer gap to the N = 50 reference (€/charger) [95 % CI]
N = 50 (continued)						
Attention+safety layer without departure bound	0.776 (0.738–0.815)	−2.01	−1.99	0.43	0	–
Attention+penalty, safety layer at deployment	0.896 (0.879–0.914)	−2.13	−2.13	0.01	0	–
Attention+penalty	1.123 (1.032–1.214)	−2.36	−2.11	0.80	0.30	–
Attention+ PPO-Lagrangian	1.187 (1.080–1.295)	−2.42	−2.33	0.19	0.46	–
Price-threshold+ safety layer	0.793†	−2.03†	−2.02†	0.12†	0†	–
ALAP+safety layer	1.038†	−2.27†	−2.27†	0†	0†	–
AFAP+safety layer	1.122†	−2.36†	−2.36†	0†	0†	–
N = 100						
SA-Attn (ours)	0.881 (0.849–0.913)	−2.16	−2.14	0.41	0	0.023 [0.012, 0.033]
Deep Sets+safety layer	0.649 (0.597–0.702)	−1.92	−1.91	0.25	0	−0.208 [−0.235, −0.182]
SA-Attn trained at N = 50 (reference)	0.857 (0.835–0.880)	−2.13	−2.10	0.45	0	–
Attention+safety layer without departure bound	0.764 (0.724–0.804)	−2.04	−2.03	0.28	0	–
Attention+penalty, safety layer at deployment	0.904 (0.866–0.943)	−2.18	−2.18	0	0	–
Attention+penalty	0.937 (0.913–0.960)	−2.21	−2.17	0.31	0.12	–
Attention+ PPO-Lagrangian	1.107 (1.065–1.150)	−2.38	−2.35	0.33	0.22	–
Price-threshold+ safety layer	0.774†	−2.05†	−2.05†	0.02†	0†	–
ALAP+safety layer	1.033†	−2.31†	−2.31†	0†	0†	–
AFAP+safety layer	1.108†	−2.38†	−2.38†	0†	0†	–
N = 200						
SA-Attn (ours)	0.902 (0.862–0.942)	−2.08	−2.04	0.55	0	0.033 [0.022, 0.046]
Deep Sets+safety layer	0.675 (0.643–0.707)	−1.85	−1.83	0.38	0	−0.194 [−0.220, −0.169]
SA-Attn trained at N = 50 (reference)	0.869 (0.838–0.901)	−2.04	−2.01	0.48	0	–
Attention+safety layer without departure bound	0.769 (0.724–0.815)	−1.94	−1.94	0.15	0	–
Attention+penalty, safety layer at deployment	0.930 (0.897–0.964)	−2.11	−2.11	0	0	–
Attention+penalty	0.929 (0.894–0.964)	−2.10	−2.10	0.08	0	–
Attention+ PPO-Lagrangian	1.119 (1.073–1.164)	−2.29	−2.23	0.63	0.78	–
Price-threshold+ safety layer	0.789†	−1.96†	−1.96†	0†	0†	–
ALAP+safety layer	1.024†	−2.20†	−2.20†	0†	0†	–
AFAP+safety layer	1.088†	−2.26†	−2.26†	0†	0†	–

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).
Transfer gap: optimality gap of the method minus that of SA-Attn trained at N = 50, paired on the shared test replays; bootstrap CI over replays. The zero-padded MLP is evaluated only for N $\leq$ 20.

uncalibrated point prediction reached a coverage of 0.899 and a gap of 1.322.

S5.5 Computation time

Figure S4 and Table S13 report the inference time per control step and the training time per run. One training run of SA-Attn took 1.15 h, and one control step took 1.68 ms at $N = 10$ and 7.76 ms at $N = 200$. At $N = 200$, Deep Sets+safety layer needed 5.99 ms per step, the rules with the layer at most 2.72 ms and eMPC-HiGHS+safety layer 14.13 ms.

Table S6: Constraint satisfaction per method over all 36 test-grid configurations (per episode): share of EVs whose stay contains no infeasible step (E4), energy not served against the shared admitted requirement, transformer limit and request pipeline.

Method	EVs without infeasible step, E4 (%)	Unserved EVs (%)	Energy not served (kWh)	Overload energy (kWh)	Episodes with overload (%)	Shortfall / deficit bound	Capped requests	Admission-reduced requests
with safety layer								
SA-Attn (ours)	84.38 (83.80–84.96)	2.24	0.43	0	0	0.009	0.52	0.13
Attention+safety layer without departure bound	100.00 (100.00–100.00)	0.99	0.09	0	0	0.007	0.52	0.13
Deep Sets+safety layer	89.98 (89.52–90.45)	1.46	0.28	0	0	0.011	0.52	0.13
MLP+safety layer	86.29 (85.15–87.43)	2.48	0.26	0	0	0.010	0.09	0.02
SA-Attn trained at N = 50 (reference)	92.94 (88.06–97.81)	0.50	0.39	0	0	0.004	0.95	0.24
LLF+safety layer	99.98†	0†	0†	0†	0†	0†	0.52†	0.13†
sLLF+safety layer	99.98†	0†	0†	0†	0†	0†	0.52†	0.13†
Price-threshold+ safety layer	95.97†	0.44†	0.03†	0†	0†	0.006†	0.52†	0.13†
AFAP+safety layer	99.96†	0†	0†	0†	0†	0†	0.52†	0.13†
ALAP+safety layer	95.83†	0.32†	0.01†	0†	0†	0.009†	0.52†	0.13†
Round Robin+safety layer	99.98†	0†	0†	0†	0†	0†	0.52†	0.13†
eMPC-HiGHS+ safety layer	84.42†	2.13†	0.45†	0†	0†	0.010†	0.52†	0.13†
SA-Attn, no admission control	84.48 (83.94–85.01)	1.98	0.40	0	0	0.008	0.52	0
Attention+penalty, safety layer at deployment	95.94 (95.22–96.65)	0.46	0.04	0	0	0.006	0.52	0.13
without safety layer								
Attention+penalty	96.63 (94.88–98.38)	1.86	0.63	0.45	6.7	–	0.52	0.13
Attention+ PPO-Lagrangian	100.00 (100.00–100.00)	0.20	0.08	0.65	7.9	–	0.52	0.13
MLP+penalty	99.79 (99.58–100.00)	1.31	1.58	1.00	17.0	–	0.09	0.02
LLF	100.00†	0†	0†	0†	0†	–	0.52†	0.13†
sLLF	99.98†	0†	0†	0†	0†	–	0.52†	0.13†
Price-threshold	73.27†	82.80†	1007.75†	25.06†	21.4†	–	0.52†	0.13†
AFAP	100.00†	0†	0†	1.25†	12.1†	–	0.52†	0.13†
ALAP	96.18†	0†	0†	0.32†	6.7†	–	0.52†	0.13†
Round Robin	100.00†	0.04†	3.0e-03†	0†	0†	–	0.52†	0.13†
eMPC-HiGHS	61.63†	99.73†	665.64†	3.7e-04†	40.5†	–	0.52†	0.13†
Attention+penalty, no admission control	96.34 (94.62–98.06)	1.55	0.47	0.42	6.5	–	0.52	0

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Shortfall / deficit bound: energy not served divided by the a-priori deficit bound of the fallback rule, averaged over the episodes with a positive deficit within each configuration and then over the configurations (methods with the safety layer). Capped: requests reduced by the arrival cap by more than 0.01 kWh. Admission-reduced: requests reduced by admission control (identical for all methods).

S5.6 Relative gaps, stress level and literature configuration

Table S14 reports relative gaps as ratios of sums for the soft-penalty optimum and for both hard optima. Because $J^{\mathrm{pen,LP}}$ is often close to zero or negative, the relative J^{pen} gap of SA-Attn exceeded 100% at $N = 5$ and $N = 10$, and the main text therefore uses the absolute gap. Table S15 covers the stress level $\rho_{\mathrm{tr}} = 3$ kW at $N = 20$ and $N = 200$. There, E4 of SA-Attn was 55.30% and 65.91%, against 65.14% and 73.25% for Deep Sets+safety layer. On EV2Gym's 25-charger configuration, Attention+safety layer without departure bound had a gap of 0.363, eMPC-HiGHS with the layer 0.456 and SA-Attn 0.507 (Table S16).

S5.7 Pooled constraint metrics

Figure S5 shows energy not served and overload energy of every method and transformer rating, pooled over all station sizes of the test grid. Overload energy appears only for the methods without the safety layer, in line with the zero overload steps of every method with the layer in Table S8. The conformal variants are not shown, and Table S12 gives their shortfall.

Table S7: E4 (share of EVs whose stay contains no infeasible step), infeasible steps, fallback activations per case and binding rates of the aggregate limits and of the departure lower bound per station size N (profiles and tightness levels pooled; mean per episode).

Method	E4 (%)	Infeasible steps	Case (i) (steps)	Case (ii) (steps)	Case (iii) (steps)	Case (iv) (steps)	Upper limit binding (% steps)	Lower limit binding (% steps)	$p_i = l_i$ (% charger-steps)
N = 5									
SA-Attn (ours)	65.86	4.64	0	0	4.64	0	4.8	0.6	42.3
Deep Sets+safety layer	76.50	3.08	0	0	3.08	0	4.6	0.4	34.1
Attention+safety layer without departure bound	100.00	0	0	0	0	0	4.2	0.3	9.3
Attention+penalty, safety layer at deployment	85.28	1.64	0	0	1.64	0	2.2	0.1	7.0
Attention+penalty	87.66	1.58	0	0	0	0	1.2	0.1	5.1
Attention+ PPO-Lagrangian	100.00	0	0	0	0	0	1.1	0	2.0e-02
MLP+safety layer	72.77	3.62	0	0	3.62	0	4.4	0.4	43.8
MLP+penalty	99.92	0.01	0	0	0	0	1.3	0	0.2
Price-threshold+ safety layer	86.89†	1.25†	0†	0†	1.25†	0†	4.4†	0.2†	31.1†
ALAP+safety layer	81.81†	1.05†	0†	0†	1.05†	0†	1.3†	0†	37.6†
AFAP+safety layer	99.75†	3.3e-03†	0†	0†	3.3e-03†	0†	2.1†	0†	0.4†
N = 10									
SA-Attn (ours)	78.64	2.50	0	0	2.50	0	2.7	0.2	37.1
Deep Sets+safety layer	86.58	1.57	0	0	1.57	0	3.0	0.2	32.1
Attention+safety layer without departure bound	100.00	0	0	0	0	0	2.5	0.1	7.0
Attention+penalty, safety layer at deployment	93.61	0.66	0	0	0.66	0	0.9	8.9e-03	4.3
Attention+penalty	94.87	0.55	0	0	0	0	0.5	8.9e-03	5.3
Attention+ PPO-Lagrangian	100.00	0	0	0	0	0	0.4	0	0.2
MLP+safety layer	87.48	1.39	0	0	1.39	0	2.2	0.1	41.1
MLP+penalty	99.45	0.05	0	0	0	0	0.4	1.7e-02	1.2
Price-threshold+ safety layer	95.19†	0.35†	0†	0†	0.35†	0†	3.0†	0.1†	29.2†
ALAP+safety layer	94.14†	0.29†	0†	0†	0.29†	0†	0.4†	0†	36.7†
AFAP+safety layer	100.00†	0†	0†	0†	0†	0†	1.0†	0†	0.1†
N = 20									
SA-Attn (ours)	86.95	1.20	0	0	1.20	0	1.6	0.1	33.3
Deep Sets+safety layer	91.97	0.73	0	0	0.73	0	2.0	0.1	31.0
Attention+safety layer without departure bound	100.00	0	0	0	0	0	1.5	2.4e-02	5.1
Attention+penalty, safety layer at deployment	97.39	0.19	0	0	0.19	0	0.3	3.7e-03	2.9
Attention+penalty	97.76	0.16	0	0	0	0	0.1	3.7e-03	5.0
Attention+ PPO-Lagrangian	100.00	0	0	0	0	0	0.1	0	0.3
MLP+safety layer	98.61	0.06	0	0	0.06	0	0.4	0	35.3
MLP+penalty	100.00	0	0	0	0	0	0.2	0	0.8
Price-threshold+ safety layer	97.52†	0.23†	0†	0†	0.23†	0†	2.3†	0.1†	29.5†
ALAP+safety layer	99.04†	0.02†	0†	0†	0.02†	0†	0.1†	0†	36.4†
AFAP+safety layer	100.00†	0†	0†	0†	0†	0†	0.4†	0†	2.4e-02†
N = 50									
SA-Attn (ours)	90.64	0.72	0	0	0.72	0	0.9	9.7e-03	30.4
Deep Sets+safety layer	94.52	0.45	0	0	0.45	0	1.5	1.2e-02	30.5
SA-Attn trained at N = 50 (reference)	92.42	0.64	0	0	0.64	0	1.0	5.2e-03	29.2

Table S7: (continued)

Method	E4 (%)	Infeasible steps	Case (i) (steps)	Case (ii) (steps)	Case (iii) (steps)	Case (iv) (steps)	Upper limit binding (% steps)	Lower limit binding (% steps)	$p_i = l_i$ (% charger-steps)
N = 50 (continued)									
Attention+safety layer without departure bound	100.00	0	0	0	0	0	0.9	2.2e-03	4.0
Attention+penalty, safety layer at deployment	99.47	0.02	0	0	0.02	0	0.1	0	1.7
Attention+penalty	99.63	0.02	0	0	0	0	0.1	0	3.6
Attention+ PPO-Lagrangian Price-threshold+ safety layer	100.00	0	0	0	0	0	0.1	0	0.6
ALAP+safety layer	98.61†	0.11†	0†	0†	0.11†	0†	1.9†	1.2e-02†	28.9†
AFAP+safety layer	100.00†	0†	0†	0†	0†	0†	1.9e-02†	0†	36.5†
N = 100	100.00†	0†	0†	0†	0†	0†	0.2†	0†	1.6e-02†
SA-Attn (ours)	91.61	0.64	0	0	0.64	0	0.8	1.6e-02	28.9
Deep Sets+safety layer	95.53	0.36	0	0	0.36	0	1.4	1.2e-02	29.7
SA-Attn trained at N = 50 (reference)	92.31	0.67	0	0	0.67	0	1.0	4.5e-03	27.5
Attention+safety layer without departure bound	100.00	0	0	0	0	0	0.8	3.0e-03	4.0
Attention+penalty, safety layer at deployment	99.86	3.3e-03	0	0	3.3e-03	0	2.4e-02	0	1.2
Attention+penalty	99.86	3.3e-03	0	0	0	0	1.9e-02	0	2.7
Attention+ PPO-Lagrangian Price-threshold+ safety layer	100.00	0	0	0	0	0	3.1e-02	0	0.9
ALAP+safety layer	97.90†	0.13†	0†	0†	0.13†	0†	1.9†	1.2e-02†	29.6†
AFAP+safety layer	100.00†	0†	0†	0†	0†	0†	3.0e-03†	0†	36.0†
N = 200	100.00†	0†	0†	0†	0†	0†	0.1†	0†	2.2e-03†
SA-Attn (ours)	92.58	0.58	0	0	0.58	0	0.8	1.2e-02	27.8
Deep Sets+safety layer	94.80	0.43	0	0	0.43	0	1.1	1.5e-03	30.0
SA-Attn trained at N = 50 (reference)	94.10	0.58	0	0	0.58	0	0.8	0	24.9
Attention+safety layer without departure bound	100.00	0	0	0	0	0	0.7	0	3.8
Attention+penalty, safety layer at deployment	100.00	0	0	0	0	0	1.5e-03	0	0.8
Attention+penalty	100.00	0	0	0	0	0	0	0	1.8
Attention+ PPO-Lagrangian Price-threshold+ safety layer	100.00	0	0	0	0	0	4.2e-02	0	1.1
ALAP+safety layer	99.70†	0.01†	0†	0†	0.01†	0†	1.3†	1.8e-02†	29.5†
AFAP+safety layer	100.00†	0†	0†	0†	0†	0†	0†	0†	36.3†
	100.00†	0†	0†	0†	0†	0†	0.1†	0†	5.1e-03†

Fallback cases: (i) contradictory limits, (ii) excess PV export, (iii) excess required power, (iv) negative budget. Limit binding: steps on which the aggregate power is at the forward or reverse limit, in % of steps. $p_i = l_i$: share of occupied charger-steps whose power equals a departure lower bound above the physical minimum.
Mean per episode of the seed-averaged policy; † deterministic method (single run).

Table S8: Event counts per method over all evaluated episodes (all test configurations, both training seeds).

Method	Episodes	Admitted EVs	Infeasible steps	Fallback activations	EVs with an infeasible step	Unserved EVs	Overload steps	Unserved EVs in episodes without infeasible step
with safety layer								
SA-Attn (ours)	7,100	233,016	17,678	17,678	39,771	6,158	0	0
Deep Sets+safety layer	7,100	233,016	12,122	12,122	27,949	5,038	0	0
MLP+safety layer	4,200	34,954	6,767	6,767	3,589	629	0	0
SA-Attn trained at N = 50 (reference)	2,700	194,466	2,689	2,689	22,583	3,643	0	0
LLF+safety layer	3,550	116,508	1	1	2	0	0	0
sLLF+safety layer	3,550	116,508	1	1	2	0	0	0
Price-threshold+ safety layer	3,550	116,508	1,675	1,675	3,025	368	0	0
AFAP+safety layer	3,550	116,508	2	2	7	0	0	0
ALAP+safety layer	3,550	116,508	936	936	1,086	84	0	0
Round Robin+safety layer	3,550	116,508	1	1	2	0	0	0
eMPC-HiGHS+ safety layer	3,550	116,508	8,595	8,595	18,765	2,526	0	0
Attention+penalty, safety layer at deployment	7,100	233,016	4,020	4,020	3,606	516	0	0
SA-Attn, no admission control	7,100	233,016	17,531	17,531	40,271	5,496	0	0
safety layer without departure bound								
Attention+safety layer without departure bound	7,100	233,016	0	0	0	1,551	0	1,551
conformal departure bound								
SA-Attn-CP (α = 0.01)	7,100	233,016	9,628	9,628	27,978	84,082	0	81,742
SA-Attn-CP (α = 0.05)	7,100	233,016	14,928	14,928	38,677	6,509	0	4,916
SA-Attn-CP (α = 0.1)	7,100	233,016	17,333	17,333	43,695	7,887	0	5,491
SA-Attn-CP (α = 0.2)	7,100	233,016	21,259	21,259	51,105	12,956	0	8,126
SA-Attn-point (uncalibrated)	7,100	233,016	17,339	17,339	43,936	7,877	0	5,469
without safety layer								
Attention+penalty	7,100	233,016	3,382	0	2,818	1,451	3,730	966
Attention+ PPO-Lagrangian	7,100	233,016	0	0	0	1,013	3,274	1,013
MLP+penalty	4,200	34,954	100	0	109	535	3,648	478
LLF	3,550	116,508	0	0	0	0	0	0
sLLF	3,550	116,508	1	0	2	0	0	0
Price-threshold	3,550	116,508	13,228	0	33,738	97,401	7,995	58,852
AFAP	3,550	116,508	0	0	0	0	2,653	0
ALAP	3,550	116,508	598	0	874	0	1,189	0
Round Robin	3,550	116,508	0	0	0	13	0	13
eMPC-HiGHS	3,550	116,508	28,719	0	48,416	116,107	16,224	60,858
Attention+penalty, no admission control	7,100	233,016	3,560	0	2,943	1,019	3,613	603

Sums over episodes. Unserved EVs: against the shared admitted requirement. Last column: episode-level check of Theorem 1(b); it is 0 whenever every unserved EV lies in an episode with an infeasible step. Theorem 1(b) covers the methods with the safety layer and the departure lower bound, not the conformal variants or the methods without the bound or the layer.

Table S9: Event counts per method and station size N (all test configurations of that N, both training seeds); columns as in the table of event counts over all episodes.

Method	Episodes	Admitted EVs	Infeasible steps	Fallback activations	EVs with an infeasible step	Unserved EVs	Overload steps	Unserved EVs in episodes without infeasible step
N = 5								
SA-Attn (ours)	1,200	3,934	5,571	5,571	1,566	382	0	0
Deep Sets+safety layer	1,200	3,934	3,691	3,691	1,100	260	0	0
Attention+safety layer without departure bound	1,200	3,934	0	0	0	130	0	130
Attention+penalty, safety layer at deployment	1,200	3,934	1,972	1,972	669	107	0	0
Attention+penalty	1,200	3,934	1,891	0	566	225	1,746	50
Attention+ PPO-Lagrangian	1,200	3,934	0	0	0	0	1,442	0
MLP+safety layer	1,200	3,934	4,346	4,346	1,236	291	0	0
MLP+penalty	1,200	3,934	8	0	4	5	1,748	3
Price-threshold+ safety layer	600	1,967	750	750	305	50	0	0
ALAP+safety layer	600	1,967	628	628	430	46	0	0
AFAP+safety layer	600	1,967	2	2	7	0	0	0
N = 10								
SA-Attn (ours)	1,200	7,798	2,998	2,998	1,879	305	0	0
Deep Sets+safety layer	1,200	7,798	1,881	1,881	1,213	195	0	0
Attention+safety layer without departure bound	1,200	7,798	0	0	0	104	0	104
Attention+penalty, safety layer at deployment	1,200	7,798	786	786	573	63	0	0
Attention+penalty	1,200	7,798	662	0	448	204	648	106
Attention+ PPO-Lagrangian	1,200	7,798	0	0	0	0	551	0
MLP+safety layer	1,200	7,798	1,664	1,664	1,111	168	0	0
MLP+penalty	1,200	7,798	60	0	56	254	609	222
Price-threshold+ safety layer	600	3,899	210	210	206	14	0	0
ALAP+safety layer	600	3,899	172	172	250	10	0	0
AFAP+safety layer	600	3,899	0	0	0	0	0	0
N = 20								
SA-Attn (ours)	1,800	23,222	5,526	5,526	5,930	1,274	0	0
Deep Sets+safety layer	1,800	23,222	4,065	4,065	4,302	977	0	0
Attention+safety layer without departure bound	1,800	23,222	0	0	0	356	0	356
Attention+penalty, safety layer at deployment	1,800	23,222	1,183	1,183	1,583	275	0	0
Attention+penalty	1,800	23,222	779	0	1,142	508	1,121	306
Attention+ PPO-Lagrangian	1,800	23,222	0	0	0	11	931	11
MLP+safety layer	1,800	23,222	757	757	1,242	170	0	0
MLP+penalty	1,800	23,222	32	0	49	276	1,291	253
Price-threshold+ safety layer	900	11,611	542	542	670	134	0	0
ALAP+safety layer	900	11,611	136	136	406	28	0	0
AFAP+safety layer	900	11,611	0	0	0	0	0	0
N = 25								
SA-Attn (ours)	200	3,596	646	646	1,871	59	0	0
Deep Sets+safety layer	200	3,596	389	389	1,002	60	0	0
Attention+safety layer without departure bound	200	3,596	0	0	0	6	0	6

Table S9: (continued)

Method	Episodes	Admitted EVs	Infeasible steps	Fallback activations	EVs with an infeasible step	Unserved EVs	Overload steps	Unserved EVs in episodes without infeasible step
N = 25 (continued)								
Attention+penalty, safety layer at deployment	200	3,596	21	21	146	0	0	0
Attention+penalty	200	3,596	10	0	100	6	53	4
Attention+ PPO-Lagrangian	200	3,596	0	0	0	4	50	4
Price-threshold+ safety layer	100	1,798	0	0	0	0	0	0
ALAP+safety layer	100	1,798	0	0	0	0	0	0
AFAP+safety layer	100	1,798	0	0	0	0	0	0
N = 50								
SA-Attn (ours)	1,200	39,014	859	859	4,093	303	0	0
Deep Sets+safety layer	1,200	39,014	541	541	2,437	225	0	0
SA-Attn trained at N = 50 (reference)	1,200	39,014	772	772	3,225	269	0	0
Attention+safety layer without departure bound	1,200	39,014	0	0	0	159	0	159
Attention+penalty, safety layer at deployment	1,200	39,014	22	22	196	3	0	0
Attention+penalty	1,200	39,014	19	0	128	303	90	301
Attention+ PPO-Lagrangian	1,200	39,014	0	0	0	75	97	75
Price-threshold+ safety layer	600	19,507	69	69	314	28	0	0
ALAP+safety layer	600	19,507	0	0	0	0	0	0
AFAP+safety layer	600	19,507	0	0	0	0	0	0
N = 100								
SA-Attn (ours)	600	39,036	382	382	3,711	186	0	0
Deep Sets+safety layer	600	39,036	214	214	1,998	118	0	0
SA-Attn trained at N = 50 (reference)	600	39,036	399	399	3,295	203	0	0
Attention+safety layer without departure bound	600	39,036	0	0	0	96	0	96
Attention+penalty, safety layer at deployment	600	39,036	2	2	69	0	0	0
Attention+penalty	600	39,036	2	0	69	111	12	111
Attention+ PPO-Lagrangian	600	39,036	0	0	0	131	19	131
Price-threshold+ safety layer	300	19,518	39	39	462	5	0	0
ALAP+safety layer	300	19,518	0	0	0	0	0	0
AFAP+safety layer	300	19,518	0	0	0	0	0	0
N = 200								
SA-Attn (ours)	900	116,416	1,696	1,696	20,721	3,649	0	0
Deep Sets+safety layer	900	116,416	1,341	1,341	15,897	3,203	0	0
SA-Attn trained at N = 50 (reference)	900	116,416	1,518	1,518	16,063	3,171	0	0
Attention+safety layer without departure bound	900	116,416	0	0	0	700	0	700
Attention+penalty, safety layer at deployment	900	116,416	34	34	370	68	0	0
Attention+penalty	900	116,416	19	0	365	94	60	88
Attention+ PPO-Lagrangian	900	116,416	0	0	0	792	184	792

Table S9: (continued)

Method	Episodes	Admitted EVs	Infeasible steps	Fallback activations	EVs with an infeasible step	Unserved EVs	Overload steps	Unserved EVs in episodes without infeasible step
N = 200 (continued)								
Price-threshold+ safety layer	450	58,208	65	65	1,068	137	0	0
ALAP+safety layer	450	58,208	0	0	0	0	0	0
AFAP+safety layer	450	58,208	0	0	0	0	0	0

Sums over episodes. The stress configurations (ρ_{tr} = 3 kW; N = 20, 200) and the literature configuration (N = 25) are included at their N.

Table S10: Cost of the guarantee per station size N (profiles and tightness levels pooled): penalty-based policy minus the same architecture with the safety layer, in J^{pen} and in raw profit, as ratios of sums over matched episodes, per admitted EV and in % of $|J^{\mathrm{LP,ceil}}|$, with the violations of the penalty-based policy.

N	Δ J^{pen} (€ per EV) [95 % CI]	Δ J^{pen} (% of $\lvert J^{\mathrm{LP,ceil}}\rvert$) [95 % CI]	Δ profit (€ per EV) [95 % CI]	Δ profit (% of $\lvert J^{\mathrm{LP,ceil}}\rvert$) [95 % CI]	Unserved EVs, penalty (%)	Overload, penalty (kWh)
Attention						
5	−3.462 [−4.639, −2.388]	−161.5 [−211.2, −113.2]	0.142 [0.075, 0.218]	6.6 [3.6, 9.9]	5.28	1.04
10	−1.407 [−1.943, −0.933]	−72.7 [−97.7, −48.7]	0.070 [0.036, 0.108]	3.6 [1.9, 5.4]	2.75	0.74
20	−0.914 [−1.224, −0.630]	−42.0 [−55.0, −29.6]	0.064 [0.039, 0.091]	3.0 [1.8, 4.1]	1.92	0.48
50	−0.311 [−0.452, −0.186]	−15.1 [−21.7, −9.3]	−0.024 [−0.042, −0.004]	−1.1 [−2.1, −0.2]	0.80	0.30
100	−0.086 [−0.132, −0.043]	−4.0 [−6.2, −2.1]	−0.057 [−0.081, −0.033]	−2.7 [−3.9, −1.5]	0.31	0.12
200	−0.042 [−0.094, 0.025]	−2.1 [−4.6, 1.2]	−0.092 [−0.120, −0.065]	−4.5 [−6.1, −3.1]	0.08	0
MLP						
5	−4.324 [−5.353, −3.283]	−201.7 [−251.6, −153.3]	−0.410 [−0.493, −0.327]	−19.1 [−23.5, −15.0]	0.09	1.80
10	−6.958 [−8.442, −5.558]	−359.3 [−449.0, −282.4]	−0.379 [−0.440, −0.318]	−19.5 [−23.0, −16.4]	2.84	0.75
20	−1.383 [−1.690, −1.104]	−63.5 [−78.6, −49.9]	−0.297 [−0.337, −0.258]	−13.7 [−15.7, −11.8]	0.99	0.44

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Δ = Σ (penalty − safety layer) / Σ denominator over the matched test replays of the seed-averaged policies; denominator: admitted EVs (identical for both policies) or $|J^{\mathrm{LP,ceil}}|$ (hard perfect-information optimum). CI: bootstrap over replays, stratified by configuration. The zero-padded MLP is evaluated only for N ≤ 20.

Table S11: Architecture and constraint handling at the training sizes N ∈ {10, 20} and in zero-shot transfer (N ≥ 50): attention, Deep Sets and zero-padded MLP policies with the safety layer, with a penalty or PPO-Lagrangian, with the layer but without the departure lower bound, and the penalty-trained policy with the layer added at deployment.

Variant	Optimality gap N ∈ {10, 20}	Unserved (%) N ∈ {10, 20}	Overload (kWh) N ∈ {10, 20}	Optimality gap N ≥ 50	Unserved (%) N ≥ 50	Overload (kWh) N ≥ 50
SA-Attn (ours)	1.132 (1.115–1.148)	2.16	0	0.901 (0.865–0.937)	0.53	0
Attention+penalty	1.887 (1.743–2.030)	2.34	0.61	0.996 (0.985–1.007)	0.40	0.14
Attention+ PPO-Lagrangian	1.525 (1.203–1.848)	0.02	0.54	1.138 (1.072–1.203)	0.38	0.49
Attention+safety layer without departure bound	0.776 (0.721–0.831)	1.14	0	0.770 (0.728–0.811)	0.28	0
Attention+penalty, safety layer at deployment	0.889 (0.879–0.899)	0.37	0	0.910 (0.880–0.940)	2.1e-03	0
Deep Sets+safety layer	0.853 (0.813–0.893)	1.31	0	0.674 (0.638–0.710)	0.37	0
MLP+safety layer	0.959 (0.944–0.975)	0.85	0	–	–	–
MLP+penalty	3.671 (2.162–5.180)	1.92	0.59	–	–	–

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).
The zero-padded MLP is evaluated only for N ≤ 20.

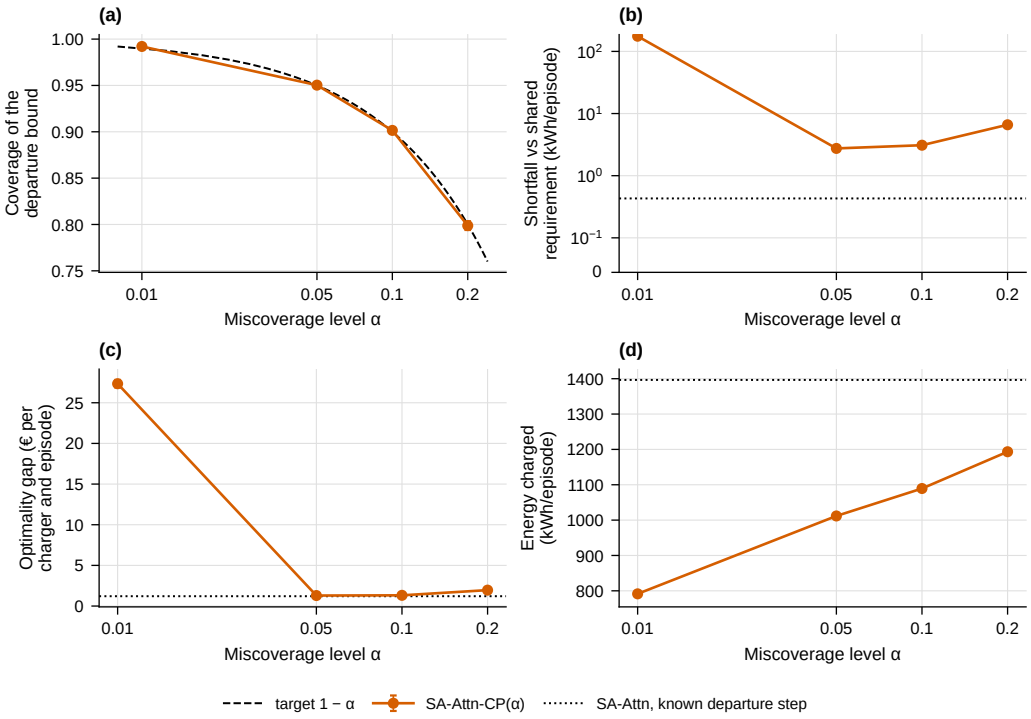


Figure S3: Trade-off of the conformal departure bound: coverage against its target, shortfall against the shared requirement, gap and charged energy per miscoverage level.

Table S12: Conformal departure bound per miscoverage level α: target and achieved coverage, shortfall against the shared admitted requirement, optimality gap and charged energy (all 36 test-grid configurations pooled; per episode unless stated).

Variant	Target coverage $1-\alpha$	Coverage (episode mean)	Shortfall vs shared requirement (kWh)	Total shortfall (kWh, all episodes)	Optimality gap (€ per charger and episode)	Energy charged (kWh)
SA-Attn, known departure step	–	–	0.43 (0.34–0.52)	2,568	1.210 (1.187–1.233)	1396.5
SA-Attn-CP (α = 0.01)	0.99	0.992	175.51 (172.55–178.46)	771,853	27.341 (27.064–27.617)	791.4
SA-Attn-CP (α = 0.05)	0.95	0.950	2.75 (2.71–2.78)	12,507	1.292 (1.290–1.294)	1011.7
SA-Attn-CP (α = 0.1)	0.90	0.902	3.09 (3.09–3.10)	13,904	1.319 (1.300–1.338)	1089.4
SA-Attn-CP (α = 0.2)	0.80	0.799	6.60 (6.50–6.70)	29,723	1.958 (1.886–2.029)	1193.4
SA-Attn-point (uncalibrated)	–	0.899	3.10 (3.09–3.10)	13,944	1.322 (1.303–1.341)	1089.1

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).
Coverage: episode mean of the fraction of EVs whose actual departure is not earlier than the conformal bound. Total shortfall: sum over all episodes of both seeds.

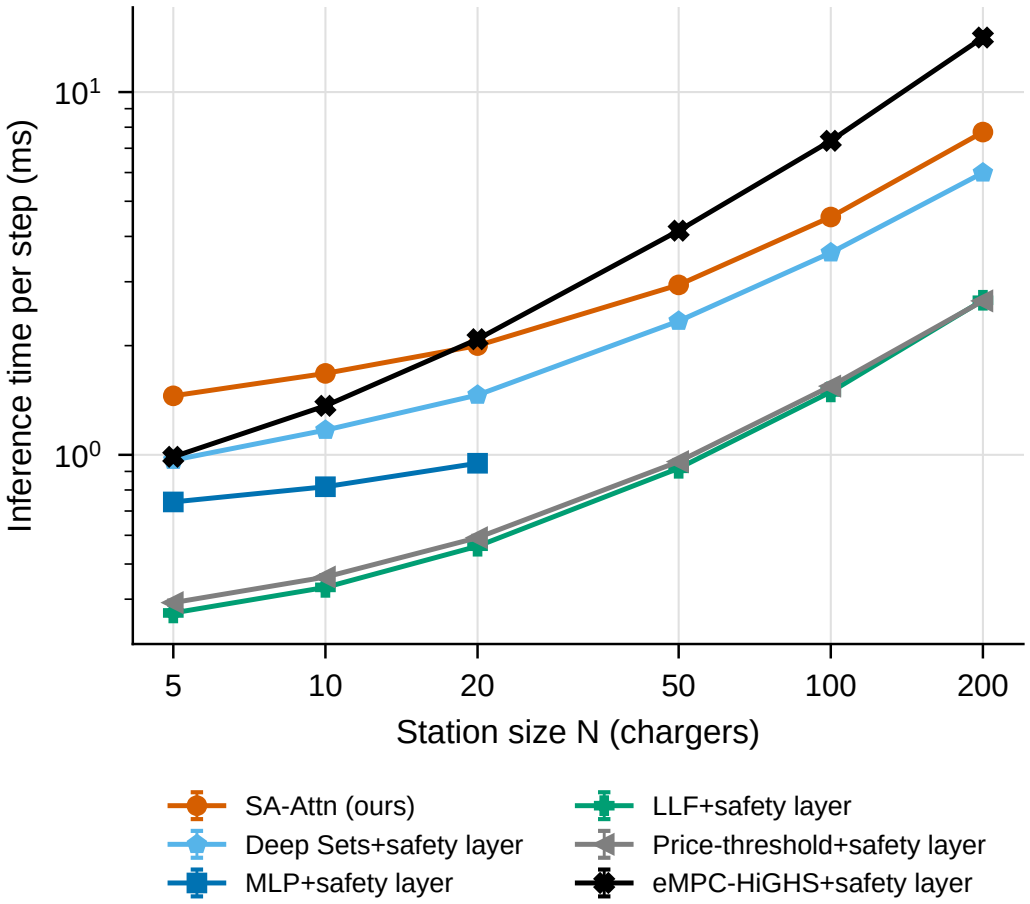


Figure S4: Inference time per control step, policy and safety layer, versus N.

Table S13: Computation: training time per run and inference time per control step (policy and safety layer) versus station size N.

Method	Training time (h)	N = 5 (ms/step)	N = 10 (ms/step)	N = 20 (ms/step)	N = 50 (ms/step)	N = 100 (ms/step)	N = 200 (ms/step)
SA-Attn (ours)	1.15 (1.14–1.16)	1.45	1.68	2.00	2.94	4.52	7.76
Attention+penalty	1.17 (1.17–1.18)	1.20	1.43	1.72	2.72	4.34	7.56
Attention+ PPO-Lagrangian	1.01 (1.01–1.02)	1.16	1.37	1.70	2.66	4.27	7.56
Attention+safety layer without departure bound	1.12 (1.11–1.12)	1.43	1.69	2.01	2.94	4.59	7.91
Deep Sets+safety layer	0.90 (0.90–0.90)	0.97	1.17	1.46	2.34	3.61	5.99
MLP+safety layer	0.81 (0.81–0.81)	0.74	0.82	0.95	–	–	–
MLP+penalty	0.79 (0.79–0.80)	0.50	0.60	0.77	–	–	–
SA-Attn trained at N = 50 (reference)	1.99 (1.98–1.99)	–	–	–	2.38	3.88	5.92
LLF+safety layer	–	0.37†	0.43†	0.56†	0.92†	1.49†	2.67†
LLF	–	0.22†	0.28†	0.41†	0.75†	1.31†	2.44†
sLLF+safety layer	–	0.36†	0.43†	0.55†	0.89†	1.49†	2.62†
sLLF	–	0.22†	0.29†	0.41†	0.75†	1.31†	2.52†
Price-threshold+ safety layer	–	0.39†	0.46†	0.59†	0.96†	1.55†	2.66†
Price-threshold	–	0.24†	0.32†	0.45†	0.79†	1.37†	2.45†
AFAP+safety layer	–	0.36†	0.43†	0.55†	0.92†	1.48†	2.64†
AFAP	–	0.20†	0.27†	0.39†	0.75†	1.33†	2.47†
ALAP+safety layer	–	0.35†	0.43†	0.55†	0.91†	1.49†	2.61†
ALAP	–	0.21†	0.28†	0.40†	0.74†	1.30†	2.40†
Round Robin+ safety layer	–	0.37†	0.43†	0.56†	0.91†	1.52†	2.72†
Round Robin	–	0.22†	0.29†	0.41†	0.76†	1.32†	2.45†
eMPC-HiGHS+ safety layer	–	0.99†	1.36†	2.08†	4.15†	7.34†	14.13†
eMPC-HiGHS	–	0.86†	1.24†	1.95†	3.96†	7.48†	14.67†
Attention+penalty, safety layer at deployment	–	1.46	1.72	2.02	2.93	4.61	7.90
SA-Attn, no admission control	–	1.37	1.64	1.94	2.89	4.46	7.71
Attention+penalty, no admission control	–	1.24	1.44	1.80	2.75	4.33	7.53

Training time: wall-clock time of each training run, mean (min–max) over the 2 runs. Inference: mean over episodes. Hardware: see the experimental setup.

Table S14: Relative optimality gaps as ratios of sums, per station size N (profiles and tightness levels pooled): $100 \cdot \Sigma(J^* - J) / \Sigma|J^*|$ over all test episodes of the seed-averaged policy, for the soft-penalty optimum (J^{pen}) and for the hard optima $J^{\mathrm{LP,ceil}}$ and J^{LP} (raw profit).

Method	J^{pen} gap vs soft optimum (%) [95 % CI]	Profit gap vs $J^{\mathrm{LP,ceil}}$ (%) [95 % CI]	Profit gap vs J^{LP} (%) [95 % CI]
N = 5			
SA-Attn (ours)	163.0 [130.4, 200.8]	56.9 [51.7, 62.5]	25.5 [22.7, 28.5]
Deep Sets+safety layer	118.3 [94.3, 146.4]	45.8 [41.8, 50.2]	16.1 [14.3, 18.0]
Attention+safety layer without departure bound	73.0 [59.3, 90.5]	49.8 [45.6, 54.5]	19.5 [17.6, 21.5]
Attention+penalty, safety layer at deployment	74.6 [65.1, 85.4]	56.8 [51.9, 62.2]	25.4 [22.9, 28.1]
Attention+penalty	324.5 [278.8, 374.0]	50.3 [45.1, 55.9]	19.9 [16.9, 23.0]
Attention+ PPO-Lagrangian	266.2 [226.5, 310.3]	73.1 [66.3, 80.6]	39.2 [35.2, 43.7]
MLP+safety layer	134.8 [107.7, 168.3]	55.1 [50.1, 60.5]	24.0 [21.5, 26.8]
MLP+penalty	336.5 [289.2, 387.7]	74.2 [67.2, 82.0]	40.2 [36.0, 44.8]
Price-threshold+ safety layer	68.0 [57.0, 81.9]	51.8 [47.1, 56.8]	21.2 [18.9, 23.8]
ALAP+safety layer	78.7 [71.2, 86.9]	69.2 [63.3, 75.7]	35.9 [32.6, 39.5]
AFAP+safety layer	73.8 [67.1, 81.5]	73.8 [67.0, 81.3]	39.8 [35.7, 44.4]
N = 10			
SA-Attn (ours)	102.5 [88.9, 118.1]	65.7 [60.7, 71.2]	27.7 [25.2, 30.3]
Deep Sets+safety layer	78.2 [65.9, 92.8]	52.2 [48.1, 56.5]	16.8 [15.3, 18.4]
Attention+safety layer without departure bound	61.7 [56.3, 67.9]	56.0 [51.8, 60.5]	19.9 [18.3, 21.6]
Attention+penalty, safety layer at deployment	72.1 [65.5, 79.7]	65.7 [60.9, 71.2]	27.7 [25.5, 30.1]
Attention+penalty	175.2 [148.7, 203.8]	62.1 [57.3, 67.4]	24.8 [22.4, 27.3]
Attention+ PPO-Lagrangian	142.4 [125.9, 160.7]	85.4 [79.0, 92.3]	43.6 [40.3, 47.3]
MLP+safety layer	81.4 [72.0, 92.3]	65.0 [60.1, 70.5]	27.1 [24.8, 29.6]
MLP+penalty	440.7 [362.2, 529.3]	84.6 [78.0, 91.7]	42.9 [39.5, 46.6]
Price-threshold+ safety layer	62.8 [57.8, 68.3]	61.2 [56.5, 66.5]	24.1 [21.8, 26.5]
ALAP+safety layer	80.0 [74.2, 86.4]	79.8 [74.1, 86.1]	39.0 [36.1, 42.2]
AFAP+safety layer	87.0 [80.4, 94.0]	87.0 [80.3, 94.1]	44.9 [41.4, 48.7]
N = 20			
SA-Attn (ours)	68.6 [63.1, 74.7]	61.4 [57.2, 66.2]	27.5 [25.5, 29.7]
Deep Sets+safety layer	50.9 [47.1, 55.2]	47.3 [44.0, 50.8]	15.9 [14.8, 17.2]
Attention+safety layer without departure bound	54.7 [51.0, 58.7]	52.2 [48.6, 56.0]	19.9 [18.6, 21.3]
Attention+penalty, safety layer at deployment	61.4 [57.3, 66.0]	61.0 [56.7, 65.5]	27.1 [25.3, 29.1]
Attention+penalty	110.6 [97.3, 125.2]	58.5 [54.4, 62.9]	25.1 [23.2, 27.1]
Attention+ PPO-Lagrangian	88.7 [81.0, 97.3]	75.7 [70.3, 81.4]	39.2 [36.5, 42.1]
MLP+safety layer	63.0 [58.7, 67.8]	63.0 [58.6, 67.7]	28.8 [26.9, 30.9]
MLP+penalty	126.6 [111.3, 143.4]	76.7 [71.2, 82.5]	40.0 [37.2, 42.9]
Price-threshold+ safety layer	55.6 [51.4, 60.0]	54.2 [50.4, 58.3]	21.5 [19.7, 23.5]
ALAP+safety layer	72.5 [67.6, 77.8]	72.5 [67.7, 77.8]	36.5 [34.2, 39.1]
AFAP+safety layer	77.5 [72.1, 83.4]	77.5 [72.0, 83.3]	40.6 [37.8, 43.6]
N = 50			
SA-Attn (ours)	68.8 [64.0, 74.4]	64.6 [60.6, 68.8]	27.4 [25.6, 29.2]
Deep Sets+safety layer	52.1 [48.4, 56.4]	49.2 [46.2, 52.4]	15.1 [14.1, 16.1]
SA-Attn trained at N = 50 (reference)	66.1 [61.6, 70.9]	62.3 [58.5, 66.3]	25.5 [23.9, 27.3]

Table S14: (continued)

Method	J^{pen} gap vs soft optimum (%) [95 % CI]	Profit gap vs $J^{\text{LP,ceil}}$ (%) [95 % CI]	Profit gap vs J^{LP} (%) [95 % CI]
N = 50 (continued)			
Attention+safety layer without departure bound	58.0 [54.5, 61.7]	56.7 [53.3, 60.3]	21.1 [19.8, 22.3]
Attention+penalty, safety layer at deployment	67.0 [63.0, 71.4]	67.0 [63.0, 71.4]	29.3 [27.5, 31.1]
Attention+penalty	83.9 [77.2, 91.5]	65.7 [61.7, 70.2]	28.3 [26.5, 30.1]
Attention+ PPO-Lagrangian	88.8 [82.5, 95.5]	81.6 [76.4, 87.3]	40.9 [38.4, 43.7]
Price-threshold+ safety layer	59.3 [55.4, 63.5]	59.0 [55.1, 63.2]	22.9 [21.1, 24.8]
ALAP+safety layer	77.6 [73.0, 82.8]	77.6 [73.0, 82.9]	37.7 [35.7, 40.0]
AFAP+safety layer	83.8 [78.5, 89.7]	83.8 [78.6, 89.7]	42.7 [40.1, 45.7]
N = 100			
SA-Attn (ours)	63.8 [58.3, 69.6]	62.4 [57.2, 68.3]	27.6 [25.1, 30.2]
Deep Sets+safety layer	47.0 [43.1, 51.5]	46.3 [42.4, 50.4]	14.6 [13.3, 15.9]
SA-Attn trained at N = 50 (reference)	62.1 [56.7, 67.9]	60.0 [55.0, 65.5]	25.7 [23.4, 28.1]
Attention+safety layer without departure bound	55.3 [50.7, 60.3]	54.7 [50.3, 59.5]	21.4 [19.6, 23.2]
Attention+penalty, safety layer at deployment	65.5 [60.2, 71.6]	65.5 [60.0, 71.6]	30.1 [27.7, 32.8]
Attention+penalty	67.8 [62.0, 74.1]	65.1 [59.6, 71.1]	29.8 [27.4, 32.4]
Attention+ PPO-Lagrangian	80.2 [72.9, 88.0]	78.0 [71.2, 85.7]	40.2 [36.8, 44.1]
Price-threshold+ safety layer	56.0 [51.1, 61.6]	56.0 [51.1, 61.4]	22.4 [19.9, 25.2]
ALAP+safety layer	74.8 [68.4, 81.8]	74.8 [68.6, 81.7]	37.6 [34.6, 40.9]
AFAP+safety layer	80.2 [73.1, 88.0]	80.2 [73.2, 88.0]	42.0 [38.4, 46.2]
N = 200			
SA-Attn (ours)	68.4 [62.2, 75.5]	65.8 [60.1, 72.1]	28.7 [26.2, 31.4]
Deep Sets+safety layer	51.2 [46.4, 56.8]	49.5 [45.4, 54.1]	15.6 [14.3, 16.9]
SA-Attn trained at N = 50 (reference)	66.0 [60.0, 72.6]	63.5 [58.2, 69.6]	26.8 [24.4, 29.5]
Attention+safety layer without departure bound	58.4 [53.5, 63.8]	58.1 [53.2, 63.4]	22.5 [20.7, 24.4]
Attention+penalty, safety layer at deployment	70.6 [64.7, 77.2]	70.6 [64.7, 77.2]	32.5 [30.0, 35.3]
Attention+penalty	70.5 [64.7, 76.9]	70.3 [64.4, 76.8]	32.3 [29.8, 35.2]
Attention+ PPO-Lagrangian	84.9 [77.3, 93.4]	80.4 [73.7, 88.1]	40.4 [37.1, 44.1]
Price-threshold+ safety layer	59.8 [54.4, 65.8]	59.8 [54.5, 65.9]	23.9 [21.4, 26.6]
ALAP+safety layer	77.7 [71.2, 84.9]	77.7 [71.2, 84.9]	38.2 [35.3, 41.4]
AFAP+safety layer	82.6 [75.4, 90.6]	82.6 [75.5, 90.4]	42.1 [38.7, 46.1]

CI: bootstrap over replays, stratified by configuration. Per-episode percentages are not reported because the optimum's J^{pen} is often close to zero or negative.

Table S15: Out-of-training transformer rating $\rho_{\mathrm{tr}} = 3$ kW per charger (stress level), per station size N (profiles pooled; per episode).

Method	Optimality gap (€ per charger and episode)	J^{pen} (€/charger)	Profit (€/charger)	Unserved EVs (%)	Overload (kWh)	E4 (%)
ρ_{tr} = 3 kW, N = 20						
SA-Attn (ours)	2.979 (2.810–3.147)	−4.25	−1.99	12.04	0	55.30
Deep Sets+safety layer	2.400 (2.288–2.512)	−3.67	−1.87	9.57	0	65.14
Attention+safety layer without departure bound	0.983 (0.938–1.028)	−2.26	−1.91	2.59	0	100.00
Attention+penalty, safety layer at deployment	1.186 (1.159–1.213)	−2.46	−2.00	2.94	0	86.09
Attention+penalty	3.183 (2.289–4.078)	−4.46	−1.97	2.67	3.94	90.08
Attention+ PPO-Lagrangian	2.243 (1.221–3.264)	−3.52	−2.21	0.04	2.62	100.00
MLP+safety layer	0.983 (0.907–1.058)	−2.26	−2.03	1.92	0	88.47
MLP+penalty	4.081 (2.983–5.179)	−5.36	−2.21	1.62	4.16	99.36
Price-threshold+ safety layer	1.077†	−2.35†	−1.95†	2.49†	0†	88.70†
ALAP+safety layer	0.916†	−2.19†	−2.16†	0.63†	0†	92.60†
AFAP+safety layer	0.950†	−2.22†	−2.22†	0†	0†	100.00†
ρ_{tr} = 3 kW, N = 200						
SA-Attn (ours)	1.586 (1.319–1.852)	−2.99	−2.20	7.21	0	65.91
Deep Sets+safety layer	1.423 (1.351–1.494)	−2.82	−2.05	6.52	0	73.25
SA-Attn trained at N = 50 (reference)	1.503 (0.971–2.035)	−2.90	−2.17	6.28	0	72.10
Attention+safety layer without departure bound	0.829 (0.755–0.903)	−2.23	−2.12	1.34	0	100.00
Attention+penalty, safety layer at deployment	0.868 (0.837–0.899)	−2.27	−2.26	0.15	0	99.09
Attention+penalty	1.021 (0.994–1.048)	−2.42	−2.26	0.08	3.15	99.11
Attention+ PPO-Lagrangian	1.441 (1.046–1.836)	−2.84	−2.41	0.71	8.20	100.00
Price-threshold+ safety layer	0.778†	−2.18†	−2.12†	0.64†	0†	95.73†
ALAP+safety layer	0.963†	−2.36†	−2.36†	0†	0†	100.00†
AFAP+safety layer	1.040†	−2.44†	−2.44†	0†	0†	100.00†

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).
The zero-padded MLP is evaluated only for N $\leq$ 20.

Table S16: Literature configuration (EV2Gym V2GProfitMax layout, 25 chargers, workplace profile), for comparison with published EV2Gym results.

Method	Optimality gap (€ per charger and episode)	J^{pen} (€/episode)	Profit (€/episode)	Energy not served (kWh)	Overload (kWh)	Inference (ms/step)
SA-Attn (ours)	0.507 (0.474–0.540)	−49.01	−47.12	0.19	0	2.11
Attention+penalty	0.706 (0.515–0.898)	−53.99	−47.83	0.01	0.60	1.85
Attention+ PPO-Lagrangian	0.790 (0.533–1.047)	−56.08	−50.32	1.8e-03	0.57	1.82
Attention+safety layer without departure bound	0.363 (0.358–0.368)	−45.42	−45.42	7.7e-04	0	2.13
Deep Sets+safety layer	0.516 (0.470–0.561)	−49.23	−46.29	0.29	0	1.59
LLF+safety layer	0.587†	−51.01†	−51.01†	0†	0†	0.60†
LLF	0.587†	−51.01†	−51.01†	0†	0†	0.46†
sLLF+safety layer	0.587†	−51.01†	−51.01†	0†	0†	0.59†
sLLF	0.587†	−51.01†	−51.01†	0†	0†	0.44†
Price-threshold+ safety layer	0.507†	−49.02†	−49.02†	0†	0†	0.64†
Price-threshold	195.993†	−4936.17†	58.80†	418.76†	80.74†	0.47†
AFAP+safety layer	0.586†	−51.00†	−51.00†	0†	0†	0.59†
AFAP	1.054†	−62.68†	−51.03†	0†	1.16†	0.45†
ALAP+safety layer	0.551†	−50.12†	−50.12†	0†	0†	0.59†
ALAP	0.589†	−51.07†	−50.12†	0†	0.09†	0.43†
Round Robin+ safety layer	0.586†	−51.00†	−51.00†	0†	0†	0.60†
Round Robin	0.586†	−51.00†	−51.00†	0†	0†	0.44†
eMPC-HiGHS+ safety layer	0.456†	−47.73†	−46.61†	0.11†	0†	2.08†
eMPC-HiGHS	104.263†	−2642.91†	−12.25†	263.07†	4.0e-04†	2.00†

Values: mean of the seed-averaged policy over all test episodes; a range in parentheses is the min–max over the 2 training seeds; † deterministic method (single run). Ranges for all columns, bootstrap confidence intervals and per-seed values are provided with the result data.
Optimality gap: J^{pen} of the soft-penalty perfect-information optimum minus J^{pen} of the method, divided by N (€ per charger and episode).

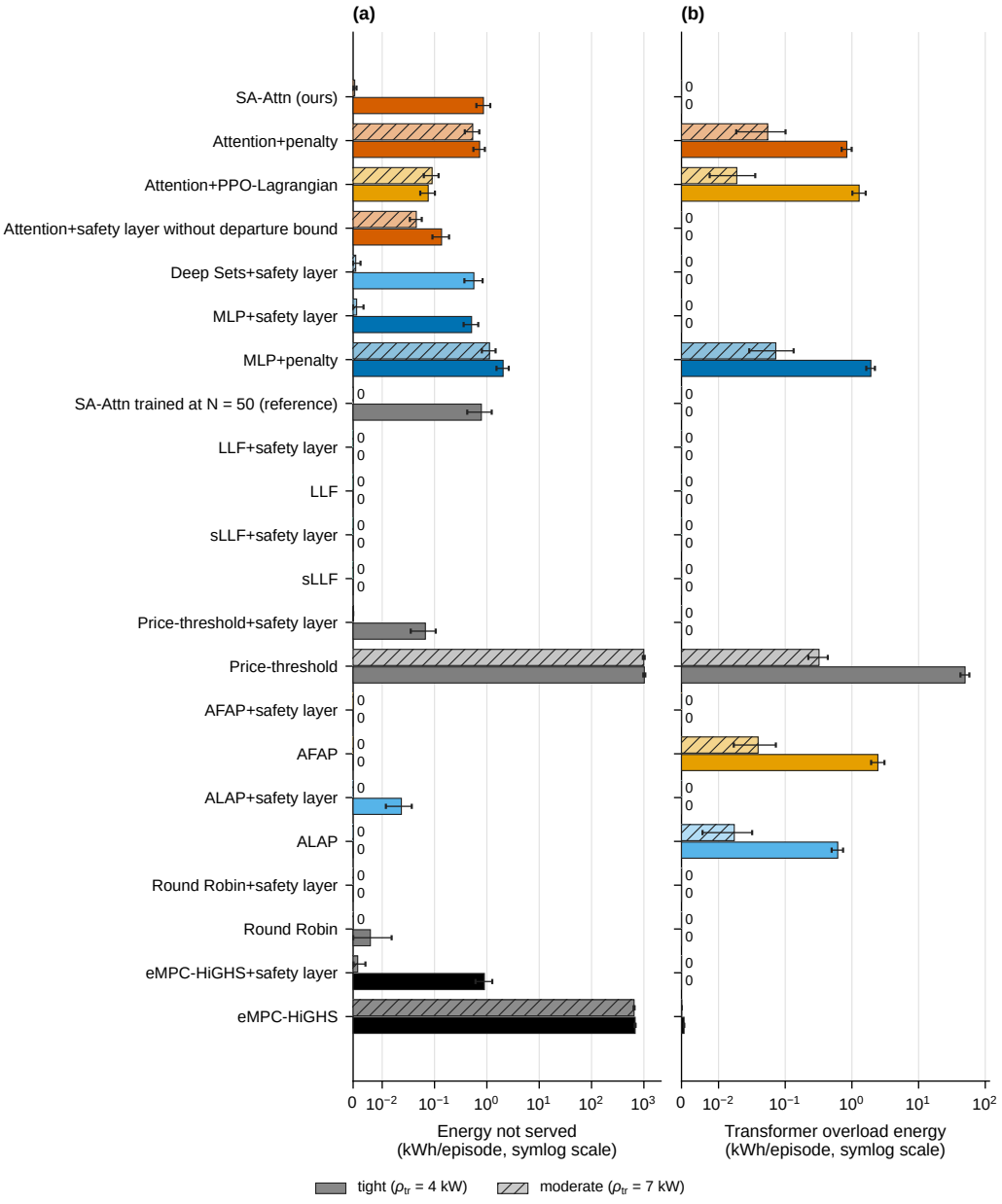


Figure S5: Constraint metrics per method and transformer rating, pooled over station sizes.

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